Low-Energy Neutrino Event Reconstruction: Clustering, Vertexing, and Energy Resolution in Phase I

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Abstract

The Deep Underground Neutrino Experiment (DUNE) aims to measure the two most energetic components of the flux of solar neutrinos (^{8}B & *hep*), as well as neutrinos from other astrophysical sources such as nearby supernovae or the diffuse supernovae neutrino background. The sensitivity to the ν-Ar charged current channel, with orders of magnitude higher cross-section than the ν elastic scattering, together with DUNE's high exposure and great energy resolution, present an exceptional advantage in understanding the characteristics of these neutrinos. This work presents enables the reconstruction of low-energy charged-current neutrino events and demonstrates the energy resolution performance in DUNE, under the presence of significant radiological and external backgrounds. Described are the development and validation of a near-threshold clustering algorithm that identifies primary ionization signals and associates adjacent energy depositions to account for spatially correlated activity. Additionally, the association of charge and light signals is described, from which the drift coordinate reconstruction can be performed.

A dedicated energy calibration strategy is employed to recover the interacting neutrino energy. The impact of various background scenarios on the reconstruction is assessed by reporting on the resolution achieved across the energy range of solar neutrino energies. Highlighted are the algorithm's efficiency in reconstructing low-energy neutrino interactions ($<$10 MeV), the accuracy of the reconstructed vertex position ($<$1 cm), and the energy resolution ($<$20 %). The results are shown to be stable across different detector geometries and background configurations. These results demonstrate the robustness of the reconstruction approach and its potential contribution to the solar-ν physics program in DUNE.

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1 Introduction

The Deep Underground Neutrino Experiment (DUNE) is a next-generation long-baseline neutrino detector. With its large 14 kt Far Detector (FD) modules [1], it will be sensitive to the two most energetic components of the solar neutrino flux (^{8}B and *hep* neutrinos), as well as, neutrinos from nearby occurring supernovae [2]. The high cross-section of Charged Current (CC) interaction on argon nuclei, together with DUNE's large active mass and excellent energy resolution, will allow for precision measurements of these astrophysical neutrino fluxes and their energy spectra. These measurements will provide deep insights into neutrino properties (the MSW effect), solar physics (the Sun's metallicity and core temperature), and potentially new physics beyond the Standard Model (non-standard interactions, sterile neutrinos, etc.).

However, the sensitivity to low-energy neutrinos in DUNE is challenged by significant radiological and external backgrounds, which will smear the signals and generate false event candidates reducing the overall sensitivity. Developing robust reconstruction techniques that can efficiently identify and reconstruct low-energy neutrino events while suppressing backgrounds is crucial for realizing DUNE's full potential in the low-energy neutrino range (5 MeV to 100 MeV).

In this work, we present a comprehensive study of low-energy neutrino event reconstruction in DUNE, focusing on clustering, vertexing, and energy reconstruction algorithms. Evaluated is the performance of these techniques for the different FD background scenarios, demonstrating their robustness for enabling precision measurements of solar neutrinos. The results presented here will be the technical foundation for the solar neutrino physics program in DUNE and will inform future analyses and detector design optimization.

1.1 The Liquid Argon Time Projection Chamber (LArTPC)

DUNE's FDs are planed to be four LArTPCs. The first two (Phase I) are already designed and the remaining (Phase II) are predicted to be one enhanced version of the Phase I modules and an additional *module of opportunity* that could be based on a different technology and/or design to optimize the sensitivity to specific physics topics.

The LArTPC technology is based on the use of liquid argon as both the target and detection medium for neutrino interactions. It was proposed by Carlo Rubbia in 1977 [3] becoming a leading technology for neutrino detection due to its excellent spatial resolution, calorimetric capabilities, and scalability. In a LArTPC, when a neutrino interacts with an argon nucleus, it produces charged particles that ionize the argon atoms along their path. The liberated electrons drift under the influence of an electric field towards a readout plane, where they are collected and recorded as signals. The readout plane is typically segmented into multiple layers (induction and collection planes) that allow for 2D reconstruction of the event. By combining the information from these planes with timing information, a full 3D reconstruction of the neutrino interaction can be achieved. Additionally, liquid argon also produces scintillation light when ionized, which can be detected by a Photon Detection System (PDS) to provide complementary information for event reconstruction and background rejection.

In DUNE, the coordinate system is designed such that the collection plane has its readout wires or strips oriented along constant z values. There are two induction planes providing useful redundancy in the plane matching to prevent the reconstruction of accidental noise coincidence. For all planes, the arrival times and charges of all signal are recorded, allowing for a projected 2D-

reconstruction of the interaction $< t, y, z >$, hence the name Time Projection Chamber (TPC). In addition to ionization signals, Liquid Argon (LAr) also produces scintillation light, which can be used for triggering, improved calorimetry, background rejection and timing (reconstructing the drift coordinate x) by implementing a PDS. For low-energy events, efficient charge collection and combined light detection and matching are crucial for a reliable 3D-reconstruction.

1.2 DUNE FDs

DUNE's FD modules are based in the Sanford Underground Research Facility (SURF) to survey a wide portfolio of neutrino physics topics. The site has 4700 m.w.e. of overburden which significantly reduces the rate of cosmic backgrounds throughout the detector and allows searches for low-energy astroparticle measurements. The PIP-II accelerator upgrade at the Fermi National Accelerator Laboratory (Fermilab) will provide a new neutrino beam directed at SURF to study long-baseline neutrino oscillations. The first Phase of DUNE will deploy two modules with approximately 14 kt of active mass. Two distinct designs (or geometries), are going to be implemented: the Horizontal-Drift (HD) and the Vertical-Drift (VD) designs (see fig. 1), named after the orientation of their drift fields.

The HD's active volume will extend 14.4 m along the drift coordinate (x), 12 m along the vertical coordinate y, and 58.2 m along the beam-aligned coordinate (z). This module will be separated into four distinct drift volumes alternating three Anode Plane Assemblys (APAs) and two Cathode Plane Assemblys (CPAs), which are set to a drift potential of 180 kV, producing an electric field of 500 V cm^{-1}. The HD's APAs are equipped with three sets of sensing wires, all with a 5 mm pitch and with induction wires oriented at ±37.5 degrees. The induction wires are positioned closest to the open drift volume, observing the signal as the free electrons transition through the layers. The rear *collection* wires absorb the charge of the drift signal and prevent the distortion of the electric field. This results in three semi-independent *views* of a neutrino interaction that can be combined to create time-dependet 2D representations of charged particle interactions in the detector. In the HD module, the two sets of induction wires wrap around the APA's centrally positioned collection wires, allowing the simultaneous observation of two drift volumes. This reduces the overall channel count but requires software intervention to deduce which drift volume an induced signal originates from. The two opposing APA sides are instrumented with their own collection plane to aid this deduction.

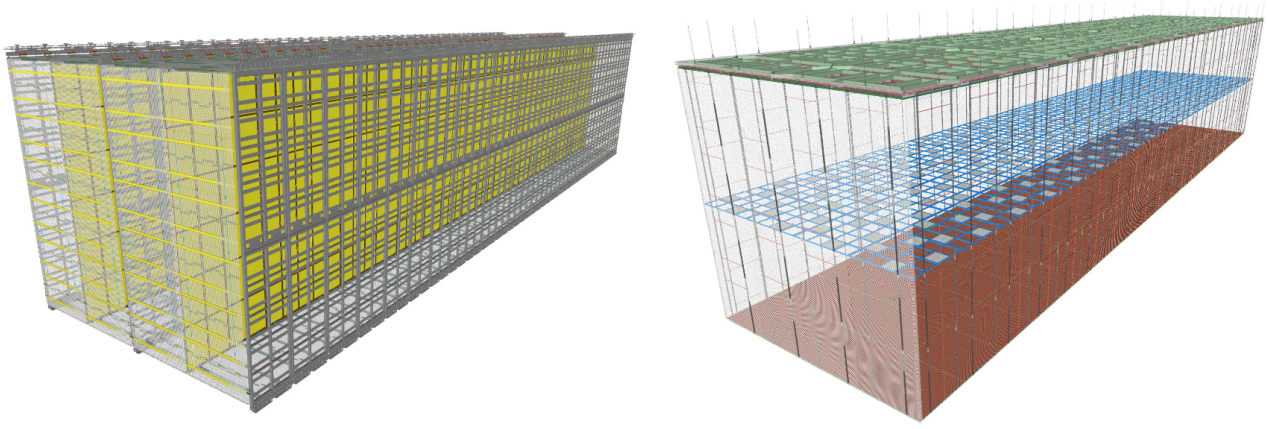


Figure 1: (Left) DUNE FD HD module. (Right) DUNE FD VD module.

The FD VD module simplifies the more traditional HD design by dividing the active volume into just two drift volumes of dimensions 13.5 m × 6.5 m × 60 m. This has been made possible by

advances in the LAr purification system that allow for longer drift distances without significant charge loss. The module is divided horizontally by the cathode, denoted simply as *Cathode Plane*. The equivalent of the CPA in the VD module is called the Charge Readout Plane (CRP) and, differently to the thin and free frame-tensioned wires of a traditional LArTPC, its building block is a perforated Printed Circuit Board (PCB) with flat readout strips (3.125 mm pitch). Similarly to the case of the free wires in the HD design, these perforations allow for the drifting electrons to pass by two induction PCB-planes, before being collected. This makes the analysis and simulation results very compatible between these two modules. The coordinate system of the VD geometry has been rotated by 90° along the z-axis.

DUNE FD Comparison

Parameter	HD Geometry	VD Geometry
LAr Mass	17.5 kt	
Inner Cryostat Volume	15.1 m × 14 m × 62 m	
Active LAr Mass	13.5 kt	14 kt
Nominal LAr Mass	10 kt	12.1 kt
Anode Modules	150 APAs	80 CRPs
Cathode Modules	300 CPAs	80 Units
Drift Length	3.6 m	7 m
Drift Field	500 V cm^{-1}	450 V cm^{-1}
TPC Voltage	−180 kV	−350 kV
TPC Views	3-view (± 37.5°)	3-view (± 30°)

Table 1: Key parameters of the HD and VD FD geometries.

Table 1 summarizes the key parameters of the HD and VD geometries. Given the nature of the LArTPC technology, in both geometries, the y and z coordinate reconstruction will follow from the TPC readout, together with a time-stamp. For beam events, the remaining x coordinate reconstruction follows from a simple calculation of the beam trigger. Note that for non-beam events (such as astrophysical neutrinos), the interaction vertex along the drift volume cannot be reliably reconstructed. To expand the reconstruction capabilities of the experiment, both detectors incorporate a PDS which makes use of X-ARAPUCA (XA) modules [4] distributed over the APA (cathode and membrane planes) for HD (VD). These provide an additional and instantaneous light signal for any interaction which, provided a correct match from the TPC readout is found, can be used to reconstruct the drift coordinate of low-energy interacting particles. This enables background fiducialisation along the drift direction.

2 DUNE's Simulation Framework

The neutrino event simulation in DUNE follows the standard low-energy workflows of the Liquid Argon Software Framework (LArSoft) Monte Carlo (MC) framework. An example of the energy deposition and charge readout of a low-energy CC neutrino interaction in LAr can be seen in fig. 2.

The simulation workflow consists of five main stages. Events are initialized using the output of the MARLEY and Decay0 particle generators, propagated through the detector using the Geometry and Tracking v.4 (Geant4) software, which are implemented within the LArSoft framework. This framework provides the necessary tools for reconstructing the events using standard and

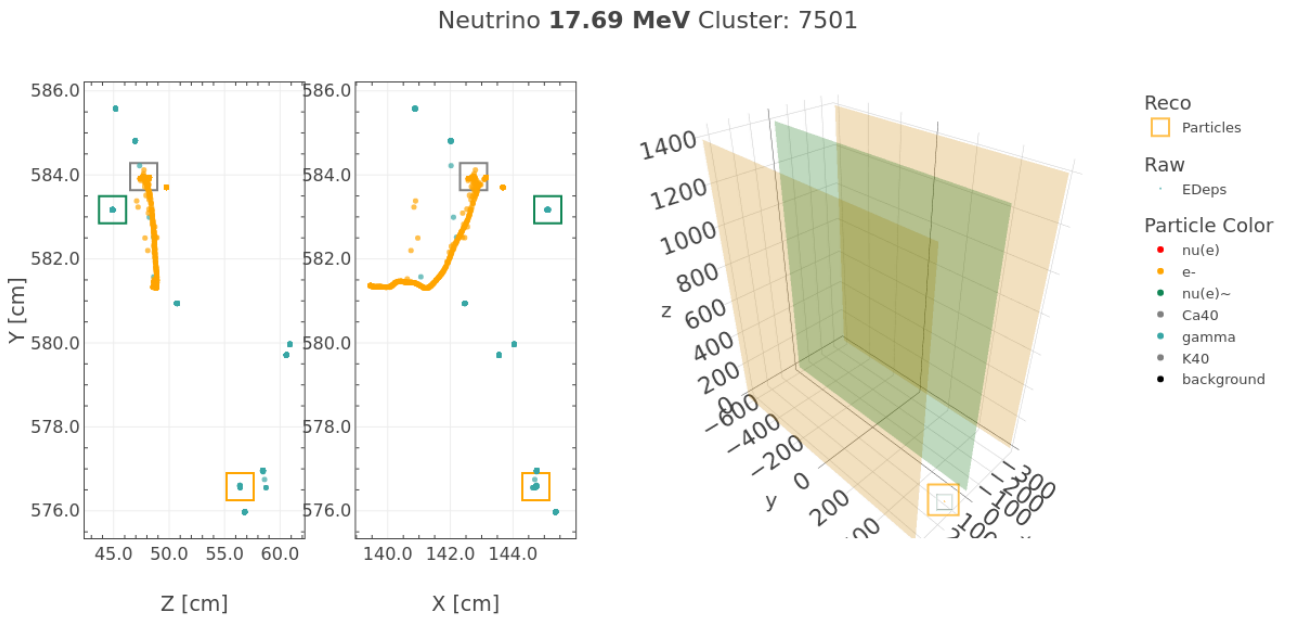


Figure 2: Example of a simulated ∼18 MeV neutrino interaction in the HD geometry. (left) The event displays the two TPC projections of the primary electron energy depositions (orange) and the associated de-excitation gammas (turquoise). The energy deposits from the truth information of the signal propagation are marked with dots and the reconstructed clusters with empty squares. (right) The reconstructed objects are displayed in a 3D representation of the workspace geometry. The represented planes are the central APA (green) and the corresponding CPAs (orange)

custom algorithms, as well as, analysis tools to output the data in a standardized format. Finally, the data is further processed with custom tools for physics extraction and results.

2.1 Workspace Geometries

Due to the high computational load of low-energy productions (including backgrounds), the full scale geometries of the FDs are represented in simulation with smaller workspace geometries; designed to provide all possible edge and propagation effects of real neutrino and background interactions but reducing the detector's volume to contain all the signal of a single neutrino event. In this work, we used two geometries: the HD workspace geometry (hd_1x2x6) and the VD workspace geometry (vd_1x8x14_3view_30deg). In both cases, the x and y coordinates are set to the same lengths as for the full drift volumes but the z coordinate is shortened roughly 1/4 and 1/3 of the full length, respectively. The HD workspace features a single central APA and two adjacent CPAs creating two drift volumes that are designed to emulate the central region of the HD detector. On the other hand, the VD workspace geometry consists of a single drift volume that represents the upper half of the detector (see fig. 3).

The HD geometry is used in low-energy neutrino studies simulating two background scenarios: one without external backgrounds from the lateral faces of the cavern APA (hd_1x2x6_centralAPA) and another one with complete external background (hd_1x2x6_lateralAPA). Due to the arrangement of the TPC planes in the workspace geometry, the latter background scenario is required to implement the lateral cavern flux in the center of the detector (where the APA is placed). This creates a scenario where the central region of the workspace actually represents the most

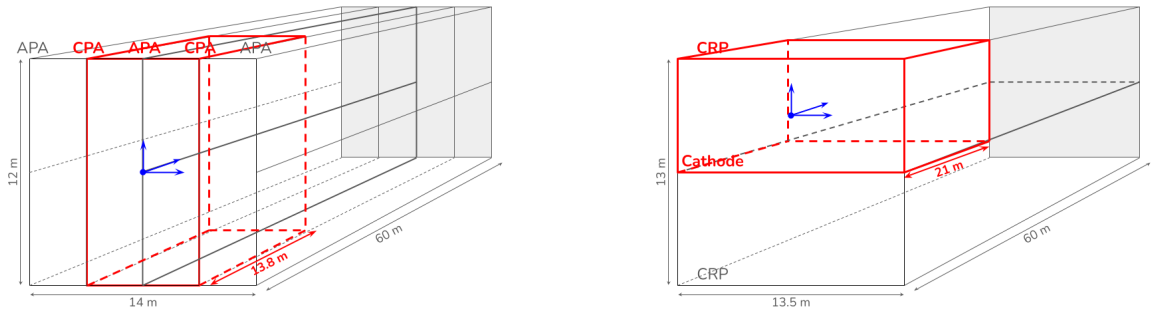


Figure 3: (left) The HD workspace geometry (hd_1x2x6_centralAPA) with a single central APA and a drift length of 3.6 m. (right) The VD workspace geometry (vd_1x8x14_3view_30deg).

external faces of the detector's charge readout. In both scenarios, we simulate the full set of radiological backgrounds, together with top, bottom, head, and tail external backgrounds. In the hd_1x2x6_lateralAPA scenario, the lateral cavern flux is also included in the central APA region.

Workspace Comparison

Parameter	HD Workspace	VD Workspace
#Drift Volumes	2	1
Active Volume	7.2 m × 12 m × 14 m	6.5 m × 13.5 m × 21 m
Active Mass	1.6 kt	2.5 kt
Anode Modules	12 APAs	28 CRPs
Cathode Modules	24 CPAs	28 Units

Table 2: Key parameters of the HD and VD workspace geometries used in the simulation.

For the VD geometry, we consider two scenarios: the nominal scenario, which represents the upper half of the VD module (vd_1x8x14_3view_30deg), and a shielded scenario that assumes additional reduction of backgrounds in the form of altered fluxes (vd_1x8x14_3view_30deg_shielded). The latter scenario is likely to represent the lower half of the VD module, where shielding is more suitable from and engineering point of view. This does not require any changes in the geometry itself, but only in the background fluxes used in the simulation. In both cases, we simulate the full set of radiological backgrounds, together with top, head, tail, and lateral external backgrounds (bottom external backgrounds are not simulated in the VD geometry due to fact that we are only simulating half of the detector's volume).

2.2 Event Generation

The simulation of low-energy neutrino interactions in DUNE relies on a comprehensive workflow that encompasses several stages, from the event generation to the final analysis data. The workflow is designed to produce realistic simulations of neutrino interactions, accounting for the complex physics processes involved and the detector response. The main stages of the simulation workflow are illustrated in fig. 4 and include:

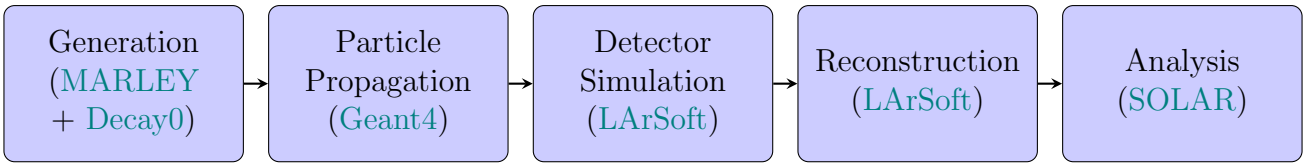


Figure 4: Simulation workflow stages for neutrino event reconstruction in DUNE. The pipeline progresses from event generation through particle propagation, detector simulation, and reconstruction to final analysis.

2.2.1 Neutrino Interactions

The first step in simulating neutrino interactions relies on a comprehensive nuclear model of the possible processes. For low-energy events, this is the MARLEY, an event generator developed to simulate the nuclear interactions of low-energy neutrinos in the range of a few to tens of MeV. It is particularly suited for simulating interactions relevant to solar and supernova neutrinos in LAr detectors. In this work we focus on the CC process (although Elastic Scattering (ES) and Neutral Current (NC) have also been implemented):

$$\nu_e + {}^{40}\mathrm{Ar} \rightarrow e^- + {}^{40}\mathrm{K}^* \tag{1}$$

Upon an electron neutrino capture on an argon nucleus and subsequent electron release, the excitation of the residual ${}^{40}\mathrm{K}^*$ leads to the emission of de-excitation γ-rays, which can produce observable signals (blips) in the vicinity (<1 m) of the main electron track (see fig. 2). MARLEY incorporates a gamma-decay schemes to describe these processes and feeds the resulting daughter particles into LArSoft, which is used to produce a MC input for physics studies. It simulates primary interaction kinematics, the resulting nuclear state and the gamma-cascades from nuclear de-excitation, as well as, other secondary processes (such as neutron or proton emission). In DUNE's physics analyses, the most important signature from these interactions is the already mentioned primary electron track. The resulting energy of these is strongly dependent on the de-excitation scheme of the residual nuclear state (see fig. 5).

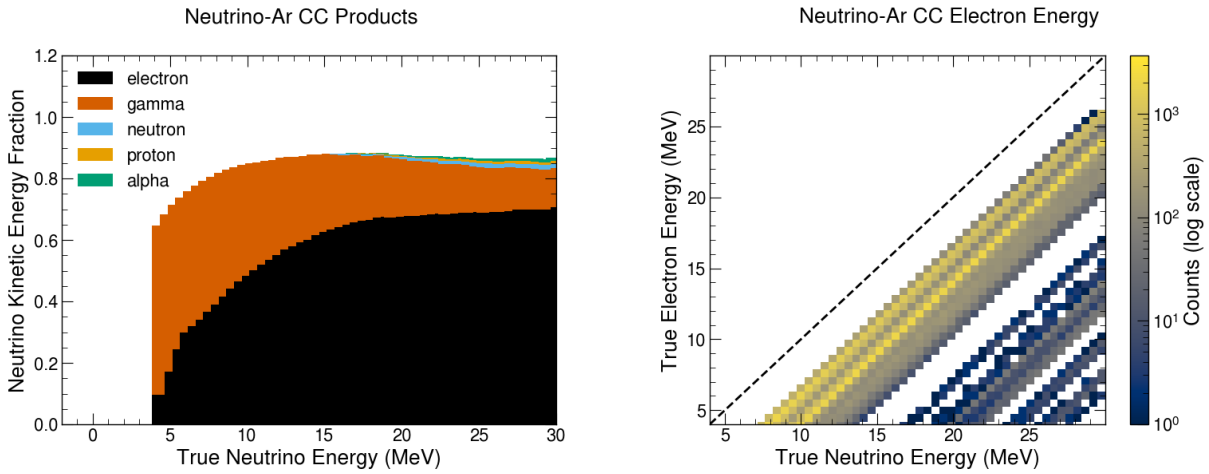


Figure 5: MARLEY's computation of CC neutrino interaction. (left) Daughter fraction of the original kinetic energy. (Right) Neutrino-electron energy correlation in a CC interaction as predicted by MARLEY.

The average energy transfer and binding energy of the CC interaction are displayed. One can observe how the gamma channel starts dominating the interaction up to ∼10 MeV, and above

that threshold, the main electron takes over the fractional energy. This trend is accompanied by a continuous increase in average kinematic threshold caused by the production of additional secondary particles. The most concerning secondary production, from the experimental point of view, is the emission of neutrons; which happens above ∼15 MeV, and leads to a decrease in visible energy that is unlikely to be recovered under realistic reconstruction scenarios. All of these effects combine to a strong energy dependence of the deposited and visible energies, which should be taken into account in our signal calibration. The same can be said about the missing energy due to nuclear binding energy losses, which are also energy dependent and lead to a non-negligible bias in the reconstructed neutrino energy if not properly accounted for. This is particularly important at higher energies, where the binding energy losses can represent a significant fraction of the total neutrino energy.

It is important to mention at this point that a typical low-energy MC production includes a full propagation of the backgrounds described in section 2.2.2. These additional particles are produced in their specific rates and concentrations according to computations out of the scope of this publication. All particle kinematics and detector placement happen in the so-called *gen* stage of the simulation. All produced background particles follow the same workflow chain as the signal particles and contribute to the ultimate smearing and resolution of the reconstructed objects. In order to account for all possible matching issues one full drift-time needs to be simulated before the neutrino makes its interaction (at $t = 0$). After this, the simulation last for another full drift-time to allow neutrinos from all regions of the detector to drift and be collected onto the corresponding readout plane. For the purpose of this publication, neutrinos are produced with isotropic positions and directions across the full active volume; and with a flat energy spectrum from 4 MeV up to 30 MeV.

2.2.2 Background Model

One of the most studied backgrounds for LAr detectors is Radon (^{222}Rn), which is a noble gas produced in the decay chain of ^{238}U [5, 6, 7]. As a gas, it can diffuse into the active volume of LAr detectors, where its decay daughters accumulate on surfaces or remain suspended, emitting beta, gamma and alpha radiation (see Appendix section 7). Alpha particles have a very short range in LAr, but due to their high energy and quenching, they produce significant light signals in the detector, which can be misidentified as low-energy neutrino interaction candidates.

The measurement of low-energy neutrinos in DUNE, particularly those from the Solar ^{8}B chain (<19 MeV), requires a thorough understanding of intrinsic (emitted from the LAr), component-induced (emanating from detector components) and external (penetrating from the cavern) backgrounds. Understanding the contributions from these particles and decaying isotopes is crucial in achieving a correct vertex and energy reconstruction required for low-energy neutrino analyses. Apart from the previously mentioned radon chain, a comprehensive list of background sources has been implemented in DUNE's simulation. Presenting DUNE's full background model is out of the scope of this work. Hence, we focus on providing a brief summary while discussing the features that have the biggest impact on the low-energy neutrino reconstruction analyses.

- **External Backgrounds**: External radiation originates outside the detector's cryostat. The flux is divided into two kinds of particles that manage to penetrate into the detector:
  - **Cavern Neutrons**: Produced by (α, n) reactions in surrounding rock or via cosmic-ray spallation, neutrons are emitted up to ∼10 MeV in kinetic energy. They can enter

the active volume where they undergo ES until they capture producing a gamma-cascade which can mimic the topology of a CC neutrino interaction. Some neutrons can be captured on the cryostat and other detector components contributing to the environmental particle flux with high energy gammas.

  - **Environmental gammas**: High-energy photons from the rock or shotcrete surrounding the detector, typically originating from natural radioactivity. The highest rate contributor to this flux in DUNE's caver at SURF is the shotcrete covering the walls. Low-energy gammas (<2.7 MeV) entering the active volume have a huge flux contributing mostly to worsen the signal reconstruction by mimicking the de-excitation gammas of the CC interaction or making the topological distinction between ES and CC less reliable. The higher energy external gammas (2.7 < E <14 MeV), are produced by γ-decay of isotopes like ^{208}Tl (∼3.5 MeV) or ^{214}Bi (∼3.27 MeV) and secondary nuclear reactions of (n, γ) interactions in certain geological environments (>6 MeV). These high-energy gamma rays are highly site-specific, depending on the local neutron flux and rock composition. Therefore, in situ measurements are fundamental to assess their impact and model accurately their contribution as fake signals to the different physics analyses.

- **Component-Induced Backgrounds**: These can arise from the materials used to construct the detector itself. Material screening and selection, as well as radon mitigation systems, are essential to reduce these sources. These include decay chains of long-lived isotopes introduced during fabrication or operation. Some of the most prominent are:

  - **Uranium (^{238}U)**: Produces alpha, beta, and gamma emissions, as well as daughters like ^{214}Bi and ^{210}Pb.
  - **Thorium (^{232}Th)**: Similar to the uranium chain, with prominent gamma emitters like ^{208}Tl (2.6 MeV).
  - **Potassium (^{40}K)**: Often present in plastics and glass (like the ones used for traditional photon detectors). It emits a 1.46 MeV gamma line.
  - **Radon (^{222}Rn)**: Present in atmospheric air and therefore very difficult to prevent from accumulation on detector and component surfaces during construction.

- **Intrinsic Liquid Argon Backgrounds**: When talking about the properties of DUNE's LAr we are generalizing the properties of (^{40}Ar) but certain radioactive isotopes are either present naturally in atmospheric argon or may be introduced as impurities:

  - **Argon (^{39}Ar)**: A beta emitter with an endpoint energy of 565 keV and a half-life of 269 years. It dominates the low-energy spectrum in atmospheric argon with an activity of $\sim 1\,\mathrm{Bq\,kg^{-1}}$. Together with the external gammas, this background contributes to the smearing of signals and worsens the hit reconstruction efficiency of the TPC by significantly altering the waveforms and providing fake signal candidates for plane-matching.
  - **Argon (^{42}Ar) and Potassium (^{42}K)**: Rare isotopes that decay via beta emission. ^{42}K emits an electron ($Q \sim 3.5$MeV), which is particularly problematic due to its similarity to low-energy solar neutrino events.
  - **Krypton (^{85}Kr)**: A beta emitter with a Q-value of 687 keV, introduced via contamination from air or during purification. Often requires cryogenic distillation to remove. It contributes to the low-energy background spectrum, similar to ^{39}Ar.

– **Radon (^{222}Rn)**: In addition to emanation from components, radon can dissolve in liquid argon and contribute to intrinsic backgrounds through its decay chain that can appear throughout the full LAr volume.

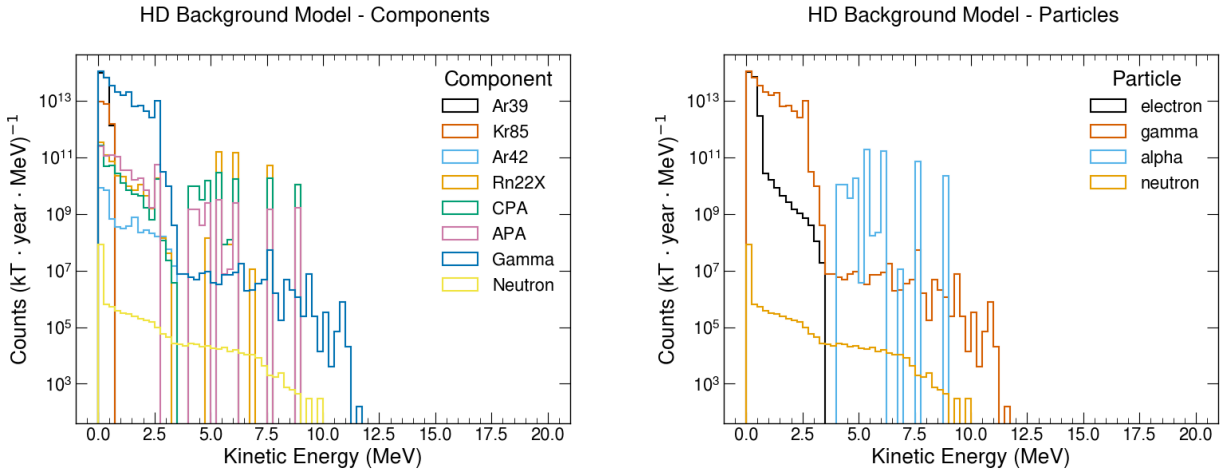


Figure 6: Simulated particle flux from DUNE's Background Model version 3. (Left) Divided by the background component. (Right) Divided by particle type. The external gamma and neutron flux are computed for the inner surfaces of the cryostat, where they are further propagated and combined with the rest of the intrinsic backgrounds.

In DUNE, LAr purification and recirculation installations are planned to ensure constant operating conditions. A detailed background model informed by in situ measurements, simulations, and radiopurity assays has been developed and is under constant revision to ensure accurate predictions of the background fluxes in the LAr active volume. The model is implemented in the LArSoft [8], the software framework used for DUNE's simulations, and is used to generate realistic event samples for detector response studies and physics analyses. Figure 6, shows the latest simulated particle flux from the background model (v. 3.6) on top of the predicted solar spectrum as a function of the particles' kinetic energy. The left panel shows the flux divided by background components, while the right panel shows it divided by particle type. The flux is dominated by external gammas, argon and krypton at low- and external gammas and neutrons at high-energies. These figures highlight the importance of understanding these backgrounds by highlighting the huge difference in flux between the low-energy neutrino signal and the background components. Without a proper energy calibration, the smearing of the background components can lead to a significant loss of signal efficiency and a degradation of the energy resolution.

2.3 Particle Propagation

Following the generation of the primary neutrino interaction via the MARLEY generator for low-energy solar neutrino events, the resulting final-state particles are propagated through the liquid argon medium using a Geant4 simulation [9]. This so called *g4* stage, models the detailed particle-matter interactions, including energy loss via ionisation, multiple scattering, bremsstrahlung, photon production, and hadronic interactions; according to a physics list tailored for these low-energy processes.

As charged particles traverse the liquid argon, they lose energy primarily through ionisation, producing free ionisation electrons and excitation of argon atoms, which subsequently decay

and emit prompt scintillation photons in the vacuum ultraviolet (VUV) range (128 nm). These processes are central to the signal generation in a LArTPC. For computational efficiency, the simulation does not explicitly track every ionisation electron or VUV photon. Instead, energy loss along particle trajectories is recorded as a set of energy depositions, or `edeps`, which include position, time, energy, and associated particle identity. Similarly, the number of scintillation photons produced in each energy deposition step is recorded as a secondary output, commonly referred to as `simphotons`. These `edeps` and `simphotons` are then effectively propagated through the LAr volume with custom analytical algorithms in a subsequent stage of simulation.

2.4 Detector Simulation

In this, so called, *detsim* stage; after the effective propagation of ionisation electrons and scintillation photons to the TPC (wires or strips) and PDS (photodetectors) readout elements, the signal formation is modelled. This stage takes as input the energy depositions (`edeps`) and the associated scintillation photons (`simphotons`) from the particle propagation step. The detector response to these inputs, produces digitized signals based on the detector's readout electronics and noise characteristics. The two components of the detector response simulation are:

- **TPC Signal**. Ionisation electrons drift under an applied electric field toward the TPC wire planes. During this drift, the electrons experience diffusion (both longitudinal and transverse), attachment to electronegative impurities (parametrised by the electron lifetime), and charge loss due to recombination. These effects are modelled to modify the total collected charge and the temporal and spatial spread of the signal. DUNEs LArTPC use three planes (also called "views") to detect the drifting charge (1 collection plane and 2 induction planes).

  1. **Induction planes** respond with bipolar signals as electrons pass by without being collected.
  2. **Collection plane** responds with unipolar signals as the electrons are finally absorbed.

  A correction is applied to account for the different responses of these planes to ensure accurate charge measurement across all views. The motion of charge induces time–dependent signals (waveforms) on the readout channels, which are digitized at high frequency. These raw waveforms are then processed using simulated electronic noise based on detector–specific measurements, which emulates realistic signal–to–noise conditions. This full set of operations are performed in LArSoft using the Wire–Cell toolkit [10]. To improve charge extraction, the waveforms undergo a **deconvolution procedure**, where the electronics response and noise spectrum are inverted in frequency space to recover a cleaner signal representation. The deconvolved waveforms are evaluated under a Region Of Interest (ROI) finder algorithm and trimmed for subsequent evaluation. This last step reduces the data volume by focusing only on regions with significant signal activity and trimming all information around the signal peak that is compatible with baseline noise.

- **PDS Signal**. Simultaneously, the scintillation photons are detected by the PDS, composed of XAs. The `simphotons` from Geant4 are converted into raw signals. Similarly to the TPC reconstruction, the PDS response is also emulated with realistic timing resolution, dark noise, and electronic gain amplification. Simulated waveforms are produced for each optical channel and overlaid with noise. Also in this case, fixed ROI windows are defined after the waveforms surpass a certain threshold, indicating potential scintillation events.

In contract to the TPC ROIs, the waveforms of the PDS can be stored and process entirely due to the limited number of channels so entire 16 µs long ROIs are stored.

With these steps completed, the simulation produces digitized waveforms for both the TPC and PDS systems, which serve as common input for the event reconstruction stage where hits and clusters are formed, as well as higher level objects for physics analyses. The output of this stage is thought to be equivalent to real detector data, allowing for direct comparison and validation of reconstruction algorithms and physics analyses.

2.5 Detector Response

The signal-processed waveforms are then passed forward to the reconstruction routines. A peak finding algorithm fits Gaussian distributions to the signal waveforms, where each successful fit is formed into an object known as a TPC *hit*. The area under all the TPC hits is proportional to the energy deposited by the charge particles as they propagate through the liquid argon volume. For the horizontal drift processing chain, a disambiguation routine is then applied to the induction plane hits to deduce which drift volume they originated from. This routine combines spatial and timing information of all hits in all planes simultaneously to maximize performance. The TPC hits are then passed forward to the higher-level reconstruction routines.

The propagation of scintillation photons in liquid argon is not performed on a per-event basis, as it would be too computationally demanding giving the high LAr light-yield, which is about 24000 photons per MeV of energy deposited for a Minimum Ionizing Particle (MIP). Instead, the propagation considering all possible effects (Rayleigh scattering, absorption by impurities, reflection and absorption on surfaces) is performed systematically in order to obtain fits used to predict the number of photons that should land on the sensitive surface of the photon detectors and their arrival time at run time. This *fast optical simulation* model is described in Ref. [11] and provides a proper dependency of the detection light-yield on the distance and offset angle between the origin of the photon and the center of a Photon Detector (PD).

Once the number of arriving photons as a function of time is obtained per PD channel, the digitization of the signal is performed. The generation of optical waveforms takes into account the expected Photon Detection Efficiency (PDE) of the XAs, the single Photoelectron (PE) signal response, Silicon Photomultiplier (SiPM) crosstalk, dark count rate, and electronic noise. Those waveforms go through a peak finding algorithm. For the VD case, this is simply done by looking for peaks above a threshold (usually corresponding to 1.5 PEs) and assigning the number of PEs as the integral of the peak divided by the area corresponding to a single PE. However, the HD case presents additional complexity, as the single PE shape shows a sizeable undershoot which requires a deconvolution to be performed before obtaining the number of PEs in a given peak.

Both reconstruction methods developed for low-energy neutrino analyses in DUNE are presented in the following chapters. Raw TPC hits, obtained after the aforementioned detector response stage, serve as the starting point for both workflows. These reconstruction objects result from the deconvolution, noise reduction and Gaussian fitting steps of the raw waveforms. Accounting for the signal's vertex of interaction within the detector volume, the TPC hit charge can be corrected, as the free electron cloud diffuses and decreases with a characteristic lifetime while propagating through the drift coordinate. This results in a lower imprint onto the TPC's collection plane depending on the signal projection distance.

Beginning at the hit level, we apply calibration steps to calculate an energy that accounts for detector response and physical processes. These include ionisation, diffusion, electron lifetime, and later, clustering losses:

- **Recombination factor**. A constant factor (0.63) is applied a priori to the integral fit of the readout waveform to account for the amount of energy that is immediately reabsorbed by the ionisation electrons by argon atoms, producing scintillation light and reducing the induced and collected charge. This value is related to the electric field strength, causing ionizing electrons to drift towards the anode.
- **Plane Calibration**. A factor is applied to each readout, U, V and W, to correct for the difference in signal current between the induced and collected charge on the final readout waveform.
- **Electron lifetime**. A correction is calculated from the exponential decay of the collected charge divided by the true electron energy as a function of the adjusted detector time and drift distance. The simulated electron lifetime is 10 ms, lower than the one measured in ProtoDUNE-HD [12].

3 Low-Energy Event Reconstruction

As a next step, after the generation, propagation and detector simulation of the MC workflow, the reconstruction (*reco*) step for these studies includes basic hit-finding algorithm and clustering schemes that applies both to the TPC and PDS. This *reco* stage typically is subdivided into two:

- *Reco1*: Conceptually, this part of the process is designed to provide the common building blocks for subsequent reconstruction stages: the `hits`. These are localized charge or photon depositions in the TPC or PDS, respectively. As quanta of information, the algorithms that produce them are optimized for efficiency and accuracy by a full team of LArSoft developers. The `hits` used in this work are the standard LArSoft objects produced from the main reconstruction workflow and common to all DUNE studies. Nevertheless, higher-level clustering algorithms have been specifically developed to group these `hits` into more complex objects that can be used for low-energy event reconstruction.
  - **Common Preprocessing**: Both TPC and PDS waveforms undergo a deconvolution step to remove the electronics response and noise spectrum. This process enhances the signal quality and prepares the waveforms for subsequent hit finding.
    * **TPC Hit Finding**: The ROI-trimmed waverforms are fitted to Gaussian Bells and integrated over time windows to form localized `hits`. These which are stored with time, position, and charge-profile information and form the fundamental building blocks for subsequent reconstruction stages.
    * **TPC Disambiguation**: This step is specifically required for HD, where induction wires turn around the central collection plane. This design allows for the reduction of the number of readout channels and simplifies, from a technical point of view, the extraction of the signals. However, it introduces an ambiguity in the hit position reconstruction, as a single readout channel can correspond to

multiple physical wire positions (left of right of the collection plane). The disambiguation algorithm uses timing and charge information from all three planes to resolve these ambiguities and assign the correct wire section to each `hit`.

* **PDS Deconvolution**: The PDS waveforms are deconvolved to remove the electronics response and noise spectrum, similar to the TPC deconvolution step. This process enhances the signal quality and prepares the waveforms for subsequent hit finding.
* **PDS Hit Finding**: The PDS deconvolved waveforms are processed to extract *optical hits* or `ophits` (integrated photon pulses) with associated time, channel and PE information. Instead of a simple Gaussian-fitting, a sliding-window algorithm is used to identify local maxima in the waveform and collect as much of the correlated scintillation light as possible into single `ophits`. This approach improves the timing and charge resolution of the detected scintillation signals.

– **Specific Low-Level Clustering**: After the `hit` finding step, specific clustering algorithms have been developed to group these low-level objects into more complex structures that can be used for low-energy event reconstruction.

* **TPC Clustering**: The TPC `hits` are grouped "per-plane" according to tight space and time constrains to form `clusters` (see fig. 7), which aim to combine all the charge from localized ionisation signals of single particles. This operation is performed for each of the TPC planes. The clustering parameters found to maximize completeness while ensuring purity in the context of background are 3 adjacent channels and 12.5 µs time window (1 tick = 12.5 µs). The resulting `per-plane clusters` contain the weighted average time and wire-position as well as the total charge.
* **TPC Matching**: The `per-plane clusters` from the three TPC planes are matched together starting from the collection plane and associating the corresponding induction plane clusters based on the shortest time (TPC-plane clocks are synchronized so that the drift time between planes is accounted for). The resulting `matched clusters` contain information from all three planes and most importantly, a reconstructed 2.5D position $< t, y, z >$.

- *Reco2*: This second part of the reconstruction process focuses on higher-level object formation and the association between TPC and PDS systems. The following operations are performed:

– **PDS Flash Finding**: The `ophits` are grouped following a simpler clustering logic with a much broader space range to account for the larger spread of scintillation light. Contrary to the TPC clustering scheme, the algorithm implemented for the PDS relies on fixed time and distance parameters that collect `adjacent ophit` surrounding a `trigger ophit` with the highest PE count. This algorithm contemplates the possibility of collecting into more than one flash the same adjacent hits. This feature can be turned on or off depending on the analysis needs. The resulting `flashes` contain time, position and total PE information for subsequent matching with TPC objects (see fig. 9).

– **Low-Energy Event Finding**: The *Reco1* TPC `cluster` are further grouped into `lowe events` containing one `primary` and any number of found `adjacent clusters` as well as a candidate list of matching PDS `flashes`:

* **TPC Adjacent Cluster Finding**: Each `cluster` is scanned for smaller `adjacent clusters` within a radial distance of 1 m (see fig. 8). The resulting `primary`

`cluster` (the highest charge cluster within 1 m) is the fundamental component to form the `lowe event`.

* **TPC-PDS Flash Matching**: Each `primary cluster` is assigned to a list of flashes via flash-matching algorithm which takes into account the reconstructed vertices of the clusters and the time and position of the flashes (find more details to the matching algorithm in section 3.3).

The clustering algorithms developed for this purpose are detailed in sections 3.1 and 3.2 and can be found in the following repositories: duneopdet & dunereco; as well as the associated producer modules: SolarOpFlash_module.cc & SolarCluster_module.cc. A custom solar workflow leading to the production of these reconstructed objects has been included in the standard production workflow files:

- solar_reco_dune10kt_1x2x6.fcl
- solar_reco_dunevd10kt_1x8x14_3view_30deg.fcl.

The calibration and characterization of the CC neutrino signal are based on the true information of the main electron of the interaction. If not otherwise mentioned, the plots in this section are produced from a MARLEY generated CC neutrino sample with central APA background. The first step of the reconstruction process has already happened when clustering the hits of the TPC and PDS detector subsystems in the simulation's *reco1* stage (see section 3.4). Next, we evaluate the detection efficiency and completeness of the collection- induction-matched TPC clusters (see **??**). Subsequently, the event electron is corrected in charge for the drift attenuation before computing a hit-dependent calibration factor that recover the particle's original kinetic energy (see section 4.2). Finally, the average calibration factor is applied to all adjacent clusters of the event candidate and the total neutrino energy is computed from four different algorithms (see section 4.4). During the TPC calibration process, the drift coordinate is assumed from truth. A discussion on how the intrinsic background contamination of the detector affects the signal reconstruction at these low energies and the effects on drift coordinate resolution (as a result of TPC-PDS match) are presented in section 5.

3.1 TPC Clustering

The clustering of TPC hits is performed using a simple algorithm based on the successive scanning of a time-ordered list of hits. The algorithm starts by taking the first hit in the list and grouping it with all hits that are within a certain distance in both channel and time. This process is repeated for each hit until all hits have been processed. The resulting clusters are then filtered based on their total charge, and only those above a certain threshold are kept for further analysis. The clustering scheme is illustrated in fig. 7. The choice of the distance threshold is crucial, as it determines the size and shape of the clusters. A smaller threshold will result in more, smaller clusters, while a larger threshold will produce fewer, larger clusters. The optimal threshold is determined based on the expected signal characteristics and the background conditions of the detector. In this work, we found that a threshold of 3 adjacent channels and a time window of 12.5 µs (25 tick) provides a good balance between cluster completeness and purity in the context of background. The resulting clusters contain the weighted average time and wire-position, as well as the total charge, which are used for subsequent reconstruction stages.

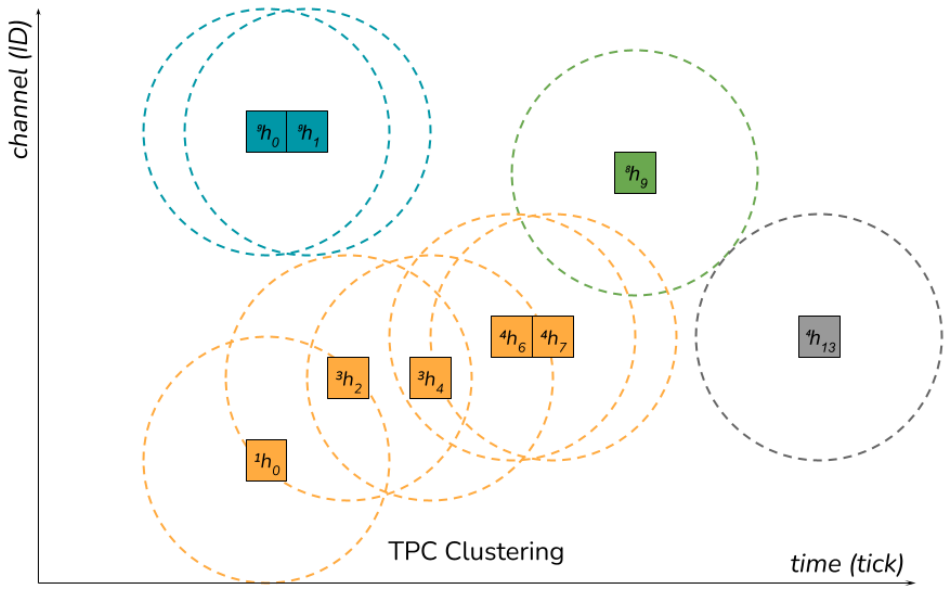


Figure 7: TPC Clustering scheme based on the successive scanning of a time-ordered `hits` (represented as black squares). The dashed circles represent the "adjacency" region for each hit. Colors are used to represent `hits` that have been grouped together. For simplicity, unitary distance in channel (ID) and time (ticks) are shown to be equivalent.

Figure 7 shows an example of the clustering process for a simulated neutrino interaction in the HD geometry. The hits are represented as black squares, and the dashed circles indicate the adjacency region for each hit. The colors represent the hits that have been grouped together into clusters. The resulting clusters contain the weighted average time and wire-position, as well as the total charge, which are used for subsequent reconstruction stages.

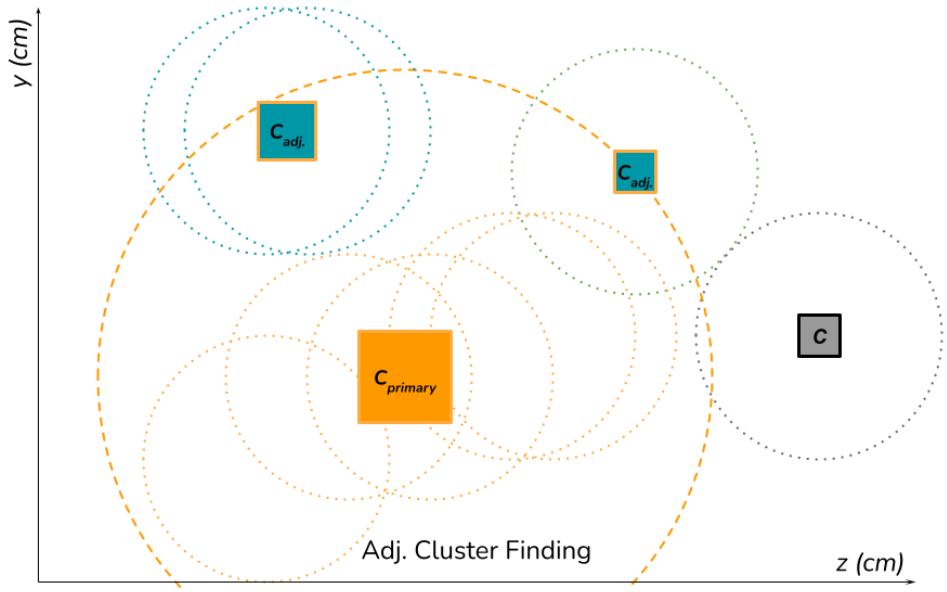


Figure 8: Adjacent cluster finding scheme. The size of the clusters represents their total charge. The dashed circle represents the 1 m radial distance used to find adjacent clusters around the primary cluster. Once found, these clusters are grouped together with the primary cluster to form a low-energy event. Clusters cannot be considered adjacent to more than one primary cluster.

Clusters are associated to others based on their relative charges and radial distance. For the radial distance comparison between clusters, the difference in their arrival times is converted to

the missing spacial coordinate by taking into account the known drift velocity in LAr. Notice that this computation only holds true for correlated energy deposits (e.g. nuclear de-excitation gammas from a CC neutrino interaction in LAr following the emission of a main electron). In the case of non-correlated background events occurring throughout the full detector volume, their distance along the drift coordinate with respect to other clusters cannot be truthfully determined. From a reconstruction point of view, this leads to the presence of a constant background floor in the vicinity of any signal cluster that corresponds to the projected activity in the interacting drift volume (see tables 5 and 6). These clusters are divided into:

- **Primary**. Clusters with charge bigger than any other's cluster charge up to 1 m radial distance. In physics studies, such as the solar neutrino analysis, these clusters can be both signal (from neutrino interactions) or background (neutrino event candidates). Here we will focus only on primary clusters from neutrino interactions, but the calibration and reconstruction techniques described here are equally applicable to background-induced primary clusters.
- **Adjacent**. Clusters with charge smaller than the one of a primary cluster closer than 1 m radial distance.

Depending on the particle producing the energy deposition, adjacent clusters can be further divided into three categories based on their origin. In the following we will distinguish between three types of adjacent clusters:

- **Signal**. These are clusters produced by secondary gammas emitted in the de-excitation of the final state nucleus after the neutrino interaction. These clusters are correlated in time and space with the primary cluster.
- **Intrinsic Background**. These are clusters produced by background sources that are intrinsic to the LAr or detector materials. These clusters are uncorrelated in time and space with respect to the primary cluster, but their rate is uniform throughout the detector volume and does not depend on the forementioned background scenarios for each of the geometries.
- **External Background**. These are clusters produced by background sources that originate outside the detector active volume (e.g. gammas from the cavern walls, neutrons from the surrounding rock, etc.). These clusters are equally uncorrelated in time and space with respect to the primary cluster and their rate depends on the background scenario for each of the geometries.

3.2 PDS Clustering

Similarly to the TPC clustering, the PDS signals are also clustered in order to reconstruct the flashes. The flash finding scheme is illustrated in fig. 9. The algorithm takes into account the time and channel proximity of the `ophits` to cluster them into flashes. Each flash is then assigned a vertex and a time, which are used for the flash-matching with the TPC clusters. The clustering is performed independently for each PDS plane in the VD geometry, while in the HD geometry all `ophits` are clustered together as there is only one PDS plane. The resulting flash contains information about the total number of PEs, the weighted average time of the flash, and

the flash vertex (computed as the weighted average position of all XA channels). The choice of the time and channel proximity thresholds is crucial, as it determines the size and purity of the flashes. A smaller threshold will result in more, smaller flashes, while a larger threshold will produce fewer, larger flashes. The optimal thresholds are determined based on the expected signal characteristics and the background conditions of the detector. In this work, we found that a time window of 100 µs and a radial distance of 1 m provides a good balance between flash completeness and purity in the context of background. The resulting flashes are then used for the flash-matching with the TPC clusters, which is described in the next section.

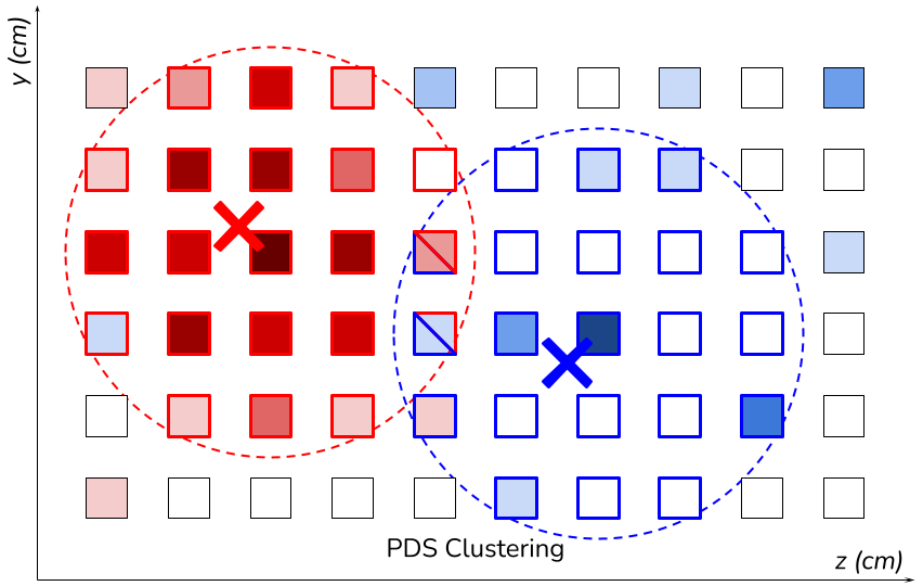


Figure 9: Flash finding scheme. Optical channels (XAs) are represented as black squares. The darkness of their filling represents the number of PEs detected per `hit`. The color represents the `ophit`'s precedence: signal (red) and background (blue). The dashed circle shows the `ophits` grouped into a single flash. Notice how the algorithm allows for the same `ophit` to be part of more than one flash.

The flashes are produced from `ophits` by clustering these PDS signals in time and channel proximity. All `ophits` within a given time window (`OpFlashAlgoMinTime` - `OpFlashAlgoMaxTime`) and distance (`OpFlashAlgoRad`) are grouped together around a single trigger `ophit`, which is defined to be the one with the highest number of PEs. The clustering is performed independently for each PDS plane in the VD detector. The resulting flash contains information about the total number of PEs, the weighted average time of the flash, and the flash vertex(computed as the weighted average position of all XA channels).

Once the set of flashes is formed, the matching is performed by evaluating the distance between the TPC vertex and the PDS flash, as well as their time difference. Flashes can be formed from signal or background activity. For any given flash, there is a non–negligible chance clustering pileup (where `ophits` formed from background activity get clustered into flashes where the trigger `ophit` originates from signal) or channel pileup (where `ophits` themselves are formed from a mixture of simultaneously arriving signal and background photons) occur. Therefore, we distinguish between:

- **Signal flashes**: those that contain at least some `ophit` originating from the signal activity. In any realistic scenario, these flashes will be contaminated with at least some amount of photons originating from background activity. Signal flashes can be subdivided into:

- **High Purity**: those that contain mostly `ophits` originating from the signal activity. Their reconstruction is likely to provide an accurate estimate of the true neutrino interaction time and good vertex reconstruction.
- **Low Purity**: those that contain mostly `ophits` originating from background activity. Their reconstruction is likely to provide a smeared estimate of the true neutrino interaction time and poor vertex reconstruction.

- **Background flashes**: those that contain only `ophits` originating from background activity. These flashes can be mistakenly matched to a TPC cluster, leading to a wrong drift coordinate reconstruction. As these are not correlated in time with the neutrino interaction, they will lead to a random and flat drift coordinate assignment within the detector active volume.

This distinction will become important as we will see in section 5.2. Before attempting any full reconstruction based on these matching, it is important to understand the reach of the matching algorithm.

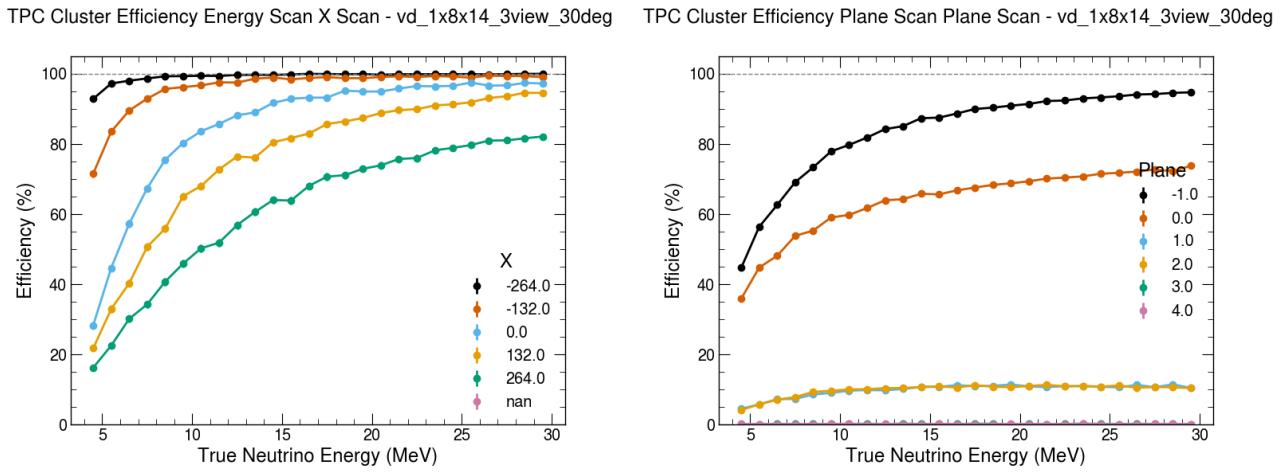


Figure 10: Flash–matching efficiency for the VD geometry. Results are shown as a function of distance to the PDS plane (left) and PDS–plane match (right).

3.3 Flash-Matching

The first step in the signal reconstruction process is the attempt to match the reconstructed primary TPC cluster with its corresponding PDS flash. Note that this step is not strictly required for the energy reconstruction, as the drift attenuation correction can be applied statistically. However, the matching is essential for the correct reconstruction of the neutrino position and, most importantly, background fiducialisation in specific physics studies. This is particularly important for low-energy solar neutrino analyses, where the signal rate is low and the high energy tails of the background spectra can leak into the signal region. The matching is also important for supernova neutrino analyses, where the time structure of the neutrino burst provides valuable information about the supernova dynamics.

The matching efficiency is defined as the fraction of TPC clusters that have been matched to a signal PDS flash. Figure 10 show the flash-matching efficiency for the HD and VD signal productions as function of the different parameters. Note that these have been used as a figure of merit to optimize the matching parameters described above. The goal is to maximize the

matching efficiency (>90%) while minimizing the reach of the clustering parameters to avoid random background matches.

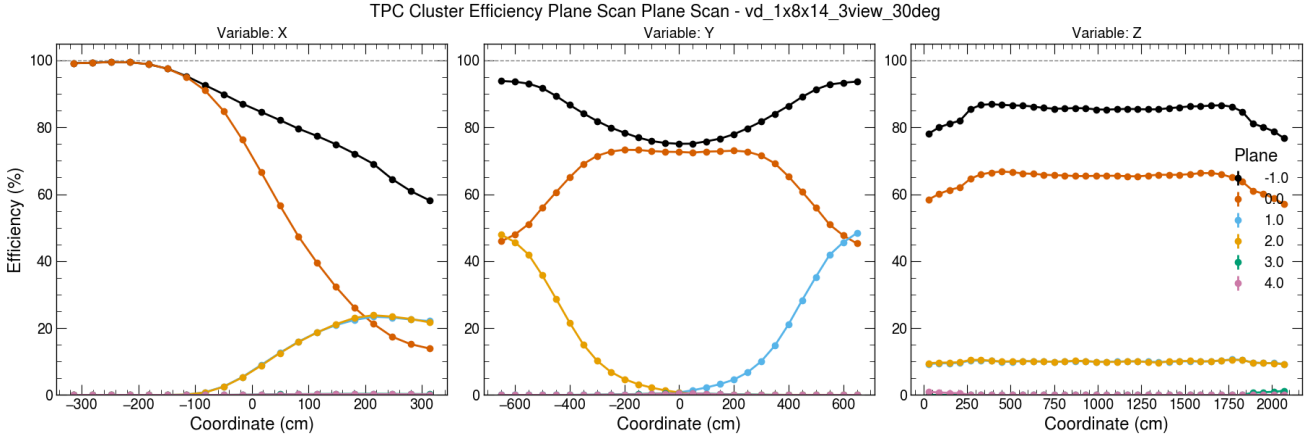


Figure 11: Flash-matching efficiency for the VD geometry. Results are shown as a function of (left) distance to the PDS plane, (center) true neutrino energy and (right) PDS plane match.

The matching efficiency grows moderately with particle energy and strongly with proximity to the PDS plane. In the case of the VD geometry, the efficiency also depends on the PDS plane being matched to, as the different planes have different PDS coverage (see fig. 11). The cathode plane has the highest efficiency, while the membrane planes closest to the anode have the lower efficiency. This is due to the fact that the light produced near the cathode plane has a higher chance of being detected by the PDS system, as it is closer to the LAr active volume.

Parameter	**Value (HD)**	**Value (VD)**
`OpFlashAlgoMinTime`	0.32 µs	0.96 µs
`OpFlashAlgoMaxTime`	0.96 µs	3.2 µs
`OpFlashAlgoRad`	300 cm	600 cm
`AdjOpFlashX`	–	140 cm
`AdjOpFlashY`	140 cm	140 cm
`AdjOpFlashZ`	140 cm	140 cm
`AdjOpFlashMinNHitCut`	0	0
`AdjOpFlashMaxPERatioCut`	1	1
`AdjOpFlashMinPECut`	0 PE	0 PE

Table 3: Flash-matching parameters for HD and VD geometries.

From this matching one can reconstruct the drift coordinate of the TPC cluster, which is essential for the correct reconstruction of the neutrino energy. The drift coordinate is computed from the time between the TPC vertex and the PDS flash by taking into account the drift velocity of the electrons in the LAr. Table 3 shows the clustering and matching parameters for each geometry used for this previous results. Note that the clustering parameters will be held the same for the different background configurations, while the matching parameters can be optimized for each configuration to maximize the matching efficiency.

For any given TPC cluster, the matching strategy follows these steps:

1. **Distance Calculation**: Compute the distance between the TPC cluster vertex and all PDS flash positions.

2. **Time Difference Calculation**: Calculate the time difference between the TPC cluster and the PDS flash. This must be smaller than the drift time of the electrons in the LAr.

3. **Candidate Selection Criteria**: A PDS flash is considered adjacent to a TPC flash if the distance $(\mathrm{TPC_y} - \mathrm{PDS_y})^2/\mathrm{AdjOpFlashY}^2 + (\mathrm{TPC_Z} - \mathrm{PDS_Z})^2/\mathrm{AdjOpFlashZ}^2$ is smaller than 1, where AdjOpFlashY and AdjOpFlashZ are the adjusted matching distances in the 2 projected coordinates, respectively. Similarly, in the case of the VD geometry matching against membrane flashes, and additional AdjOpFlashX parameter is used where the $(\mathrm{TPC_X} - \mathrm{PDS_X})^2/\mathrm{AdjOpFlashX}^2$ term is computed from the time difference between the TPC cluster and the PDS flash.

4. **Matching Criteria**: The algorithm consider 3 additional cuts to prevent the random matching of background cluster:

   - The minimum number of hits in the PDS flash: `AdjOpFlashMinNHitCut` (by default 3).
   - The maximum ratio of `AdjOpFlashMaxPE` / `AdjOpFlashPE`: `AdjOpFlashMaxPERatioCut` (by default 1). This is intended to prevent the matching of background PDS flashes, which are less homogeneous in the number of PEs compared to the signal PDS flashes.
   - The minimum & maximum number of PEs `AdjOpFlashMinPECut` & `AdjOpFlashMaxPECut`. This cut is tuned to match the expected light-map of a neutrino production.

This algorithm reduces the initial selection of PDS flashes to a set of matching candidates. From this set, only one flash is selected as the match for the TPC cluster. To do so, a light-map based parametrization has been implemented by evaluating the expected number of PEs in the flash as a function of the distance to cluster's charge and predicted drift distance. Hence, the algorithm picks out the flash candidate that minimizes the following quantity:

$$\chi = \frac{\mathrm{abs}(\mathrm{PE_{flash}} - \mathrm{PE_{expected}})}{\mathrm{PE_{flash}^{\alpha}}}\,, \tag{2}$$

where $\mathrm{PE_{flash}}$ is the number of PEs in the flash, $\mathrm{PE_{expected}}$ is the expected number of PEs from the light-map parametrization, and α is a tunable parameter (by default set to 2) that controls the weight of the flash PE count in the matching.

This parametrization is helpful to investigate which matching strategy works best:

- **Minimum Error Matching** ($\alpha = 0$): select the flash candidate that minimizes the absolute difference between the number of PEs in the flash and the expected number of PEs from the light-map.
- **Minimum Relative Error Matching** ($\alpha = 1$): select the flash candidate that minimizes the relative difference between the number of PEs in the flash and the expected number of PEs from the light-map.
- **Weighted Relative Error Matching** ($\alpha = 2$): select the flash candidate that minimizes the weighted relative difference between the number of PEs in the flash and the expected number of PEs from the light-map.
- **Maximum PE Matching** ($\alpha \to \infty$): select the flash candidate with the highest number of PEs.

TPC–PDS Matched Neutrino Sample (%)				
Configuration	No Match	Background	Low–Purity	High–Purity
HD Signal	1.72(2)	—	—	98.3(2)
HD Central	0.56(3)	39.0(2)	1.29(4)	59.1(3)
HD Lateral	0.56(3)	34.8(2)	0.44(2)	64.2(3)
VD Signal	12.32(7)	—	—	87.7(2)
VD Nominal	5.9(2)	54.5(7)	34.6(5)	5.0(2)

Table 4: TPC-PDS matching comparison between geometries. Fraction of sample matched to flash.

In the following we detail the implementation of the light-map assisted matching method. This algorithm takes into account the expected light yield distribution for a given signal TPC cluster:

$$\log_{10}(\mathrm{PE}) = A - A \cdot \lambda \cdot \frac{t}{t_{max}} + C \cdot \left(\frac{t}{t_{max}}\right)^2 . \tag{3}$$

Where A, λ and C are amplitude, attenuation, and correction fit parameters; t is the drift time and t_{max} is the maximum drift time in the detector. This formula can be used to extract different attenuation curves for different cluster charge ranges that can be fitted to simple quadratic functions in the logarithmic space to parametrize the light-map assisted PE cut as a function of both drift time and cluster charge Q:

$$\mathrm{PE}_{\mathrm{cut}}(t, Q) = \exp A(Q) - A(Q) \cdot \lambda(Q) \cdot \frac{t}{t_{max}} + C(Q) \cdot \left(\frac{t}{t_{max}}\right)^2 , \tag{4}$$

with $A(Q)$,$\lambda(Q)$, $C(Q)$ fitted with a quadratic function of Q

$$\alpha(Q) = a_0 + a_1 \cdot Q + a_2 \cdot Q^2 \quad \text{where } \alpha \in \{A, \lambda, C\} . \tag{5}$$

Where $A(Q)$, $\lambda(Q)$ and $C(Q)$ are amplitude, attenuation, and correction fit parameters as a function of the cluster charge Q; t is the drift time and t_{max} is the maximum drift time in the detector.

In terms of flash-matching efficiency, we can see in table 4 that the HD geometry has a better overall performance than the VD geometry, which is expected due to the larger drift volume of the latter. The vd_1x8x14_3view_30deg_nominal configuration has the lower matching efficiency due to the larger drift volume and higher background rate, which increases the probability of miss-matching.

3.4 Event Definition

The first figure of merit to be extracted from the simulation is the detection efficiency (see fig. 13 (left)). We observe a strong dependence on the number of hits. Only with a minimum hit condition of 2, the simulation reaches detection efficiency values of ∼100%. Achieving a reconstruction threshold (>90% efficiency) above 5.5 MeV requires a minimum hit condition of 1 hit. Instead of interpreting this efficiency in terms of the neutrino electron, using the information provided

in ?? (neutrino–electron energy offset >4.5 MeV) one can deduce an electron detection threshold of ∼1 MeV. A better estimation of the detection threshold can be extracted from the energy resolution fit in ??.

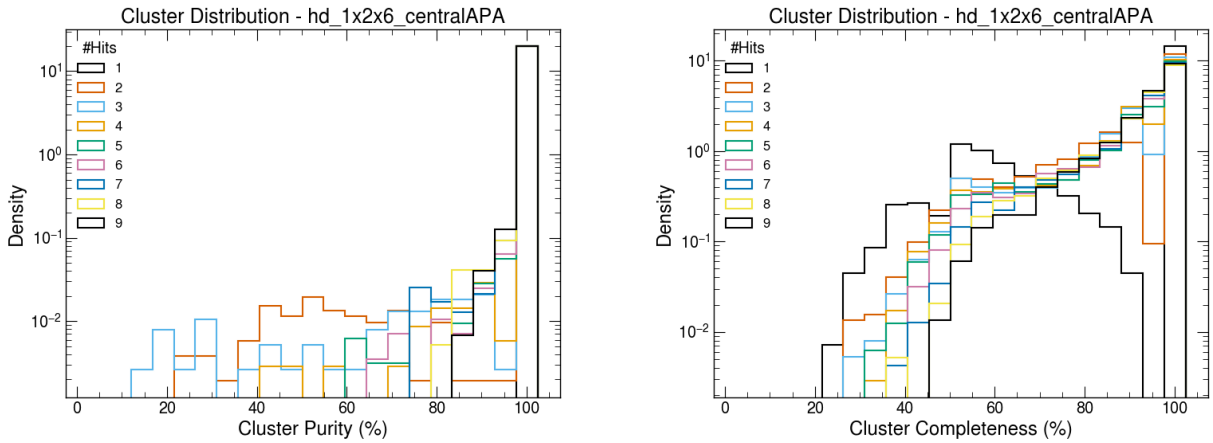


Figure 12: Clustering efficiency plots. (left) Neutrino signal purity in the main cluster. (right) Electron charge completeness in the main cluster.

The next parameter to evaluate is the completeness of the main clusters, which is a measure of the electron charge–confinement given a certain clustering parameter set. For the purpose of the solar neutrino analysis of DUNE, clustering parameters of 3 adjacent channels and 25 ticks have been used to maximize completeness while ensuring cluster purity in the context of background. The most up-to-date values for this clustering should be found in the producer's SolarCluster.fcl. Despite the good overall completeness of the signal, we observe a trend with respect to the number of hits per cluster, which can be accounted for in the later described charged correction factor to improve reconstruction. The trend that one observes is clear and understandable; for depositions which result in a lower number of hits (this can happen because of low energy deposition or a perpendicular trajectory of the particle with respect to the TPC plane) it is more likely that clusters of similar sizes are formed (e.g. 2 clusters of 1 hit) resulting in a lower charge in the main cluster.

DUNE's solar analysis workflow starts with calibrating the clustered hits from the CC interaction of simulated neutrino events with the MARLEY generator. The CC channel is chosen to be the default based on its dominant cross-section in the low-energy range, as well as its richer topology. All energy and spacial resolution results presented here should be considered an underestimation (or at least compatible) to the ones results for the ES channel. Each cluster's information is reduced to a point-like object characterized by its number of hits (`nhits`), `charge` (summed over the charge of the individual hits) and a 2.5D-reconstructed vertex $< t, y, z >$ computed from the charge-weighted average of the individual hits.

We define a low-energy event as the combination of a `primary cluster` together with its `adjacent clusters` and a list of `flashes` that are matching candidates for the drift coordinate reconstruction (see section 3.3). In any neutrino analysis it will become important to try to distinguish (at least on average) the `adjacent cluster` distributions for any low-energy event, not only to distinguish background activity (e.g. neutron captures) from signal events, but also to identify the different interaction channels of the neutrino interaction (CC events produce de-excitation gammas that are not present in ES interactions). Consequently, the higher the projected background activity in any of the isolated drift volumes of DUNE's FDs, the harder particle identification and channel separation will become. Notice that there is nothing

Adjacent Cluster Distribution - Vertex Distance 20 cm

Configuration	Source		
	Signal	Intrinsic	External
HD Signal	0.759 ± 0.002	—	—
HD Lateral	0.609 ± 0.002	0.0165 ± 0.0003	0.0228 ± 0.0003
HD Central	0.538 ± 0.002	0.0181 ± 0.0002	0.0131 ± 0.0002
VD Signal	0.890 ± 0.002	—	—
VD Nominal	0.503 ± 0.004	0.147 ± 0.002	0.224 ± 0.002
VD Shielded	0.531 ± 0.004	0.145 ± 0.002	0.0518 ± 0.0009

Table 5: Average number of adjacent clusters per primary signal cluster for 20 cm distance.

Adjacent Cluster Distribution - Vertex Distance 100 cm

Configuration	Source		
	Signal	Intrinsic	External
HD Signal	1.277 ± 0.003	—	—
HD Lateral	1.048 ± 0.003	0.2094 ± 0.0010	0.311 ± 0.001
HD Central	0.941 ± 0.003	0.234 ± 0.001	0.1739 ± 0.0010
VD Signal	1.489 ± 0.004	—	—
VD Nominal	0.988 ± 0.007	2.37 ± 0.01	3.43 ± 0.02
VD Shielded	1.017 ± 0.006	2.30 ± 0.01	0.778 ± 0.005

Table 6: Average number of adjacent clusters per primary signal cluster for 100 cm distance.

preventing a single neutrino interaction from producing more than one low-energy event. Rarely, but not inevitably, secondary gammas scatter further than 1 m in LAr or a background-induced miss-match in the TPC reconstruction leads to blips appearing in uncorrelated regions of the detector. Independently of this possibility, a low-energy event reconstruction efficiency can be defined as the possibility of a single particle interaction to produce at least one low-energy event that can be fully reconstructed. This efficiency is shown in fig. 13 for both HD and VD geometries as a function of the cluster's `nhit` threshold value. The chosen signal clustering parameters are 3 adjacent channels and 12.5 µs time window. The results show a clear and expected decrease in efficiency with increasing `nhit` threshold, as low-energy events are more likely to be lost when applying stricter cuts on the minimum number of hits per cluster. This shows the importance of a simple clustering approach to low-energy neutrino reconstruction, as more sophisticated algorithms might lead to a considerable decrease in efficiency.

For the purpose of this work, secondary low-energy events coming from a single neutrino interaction will always be ignored in favor of the one with the highest charge in the primary cluster. This is to ensure that the reconstruction performance is evaluated for the most complete reconstruction object in any interaction but in any physics analysis these should be kept and treated as part of the signal or even background samples.

For the specific purpose of the solar neutrino studies an analyzer module (SolarNuAna_module.cc) has been written to combine these reconstruction objects with the truth information of the event. This module essentially combines all the information needed to perform the calibration and reconstruction of the signal candidates. The biggest caveat from the analysis point of view is that all clusters and flash information is reduced to a point-like data structure. This is, of course,

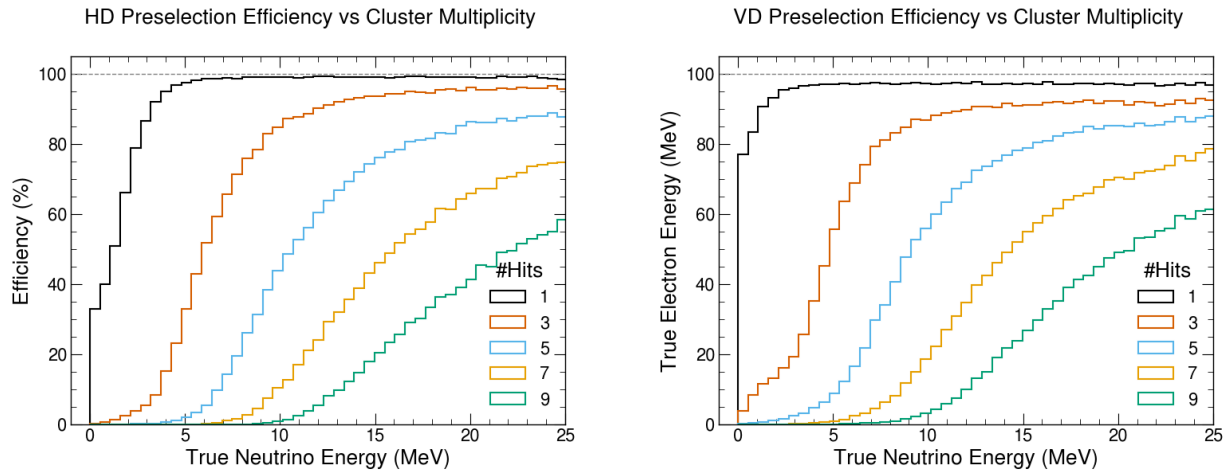


Figure 13: HD (left) and VD (right) detection efficiency for signal clustering parameters of 3 adjacent channels and 12.5 µs.

a choice of design in the analyzer and allows for a simpler handling of the data because all event candidates have the same dimension. In future works, this analyzer should be substituted by the more general LowEAna_module.cc which aims to provide the same event reconstruction while keeping the internal structure of the candidate clusters to allow for better calibration and discrimination strategies.

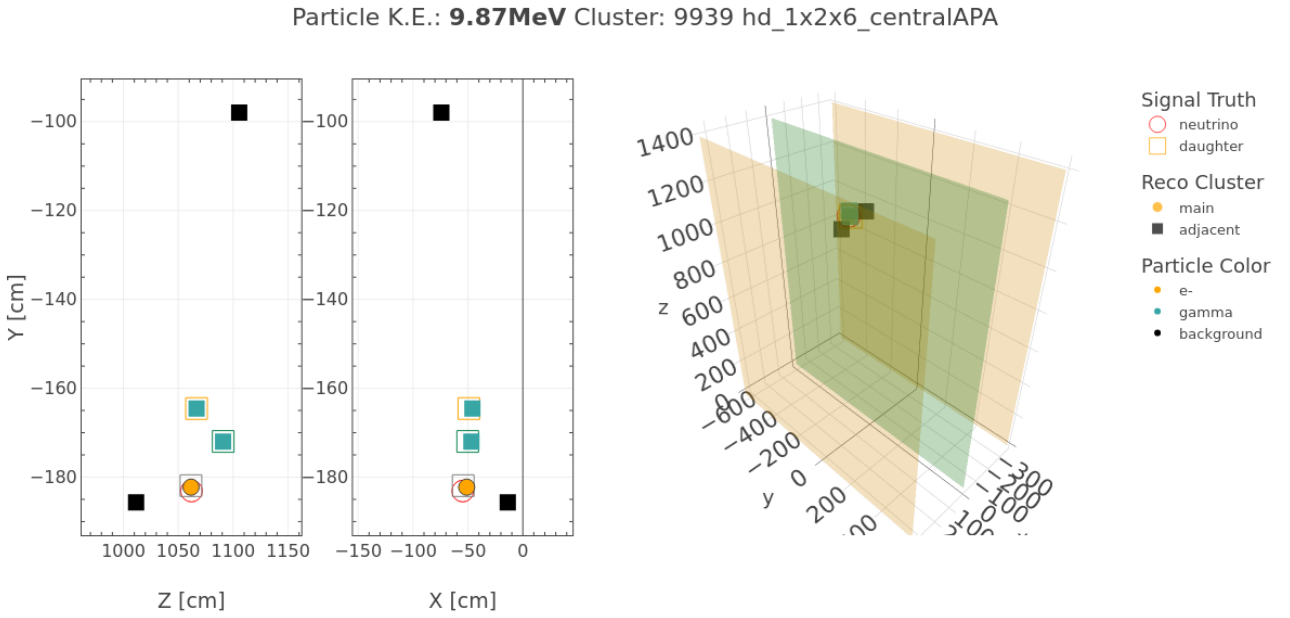


Figure 14: Example of a simulated ∼10 MeV neutrino interaction in the HD geometry. (left) The event displays the two TPC projections of the primary electron cluster (orange), the associated de-excitation gammas (turquoise) and uncorrelated background activity (black). The biggest energy deposits from the truth information of the signal propagation are marked with empty squares and the reconstructed clusters with filled squares. (right) The same points are displayed in a 3D representation of the workspace geometry. The represented planes are the central APA (green) and the corresponding CPAs (orange)

An example of a simulated neutrino interaction in the HD geometry is shown in fig. 14. The event displays the two TPC projections of the primary electron cluster (orange), the associated de-excitation gammas (turquoise) and uncorrelated background activity (black). The biggest

energy deposits from the truth information of the signal propagation are marked with empty squares and the reconstructed clusters with filled squares. Compared to fig. 2, the clusters are now reduced to a point-like data structure, which results from the clustering algorithm.

4 Neutrino Energy Calibration

Once the reconstruction algorithms are defined and we have obtained the raw variables from the TPC and PDS subsystems, we can proceed to calibrate the neutrino energy. The calibration process is based on the true information of the main electron of the interaction, which allows us to compute a hit-dependent calibration workflow that recovers the particle's original kinetic energy. The average calibration factor is then applied to all adjacent clusters of the event candidate and the total neutrino energy is computed from four different algorithms (see section 4.4). During the TPC calibration process, the drift coordinate is assumed from truth. A discussion on how the intrinsic background contamination of the detector affects the signal reconstruction at these low energies and the effects on drift coordinate resolution (as a result of TPC–PDS match) are presented in section 5.

4.1 Electron-Lifetime Correction

TPC hits are first grouped into spatial clusters using a dedicated clustering algorithm. Each event contains a **main cluster**, corresponding to the primary outgoing electron, and zero or more **adjacent clusters**, interpreted as gamma-induced "blips". The first step in calibration involves correcting the total ionization charge of each cluster for the electron-lifetime attenuation in the LAr (see section 4.1).

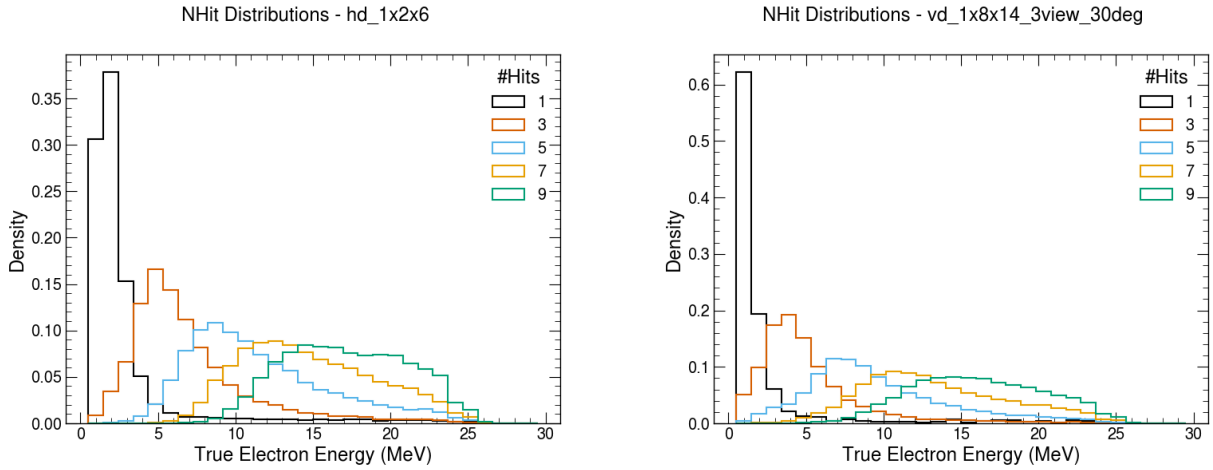


Figure 15: Neutrino energy correlation with number of hits in the primary cluster.

$$\frac{Q_{\text{raw}}(n_{\text{hit}})}{E_e^{\text{truth}}} = F_0(n_{\text{hit}}) \cdot \exp\left(\frac{t^{\text{drift}}}{\tau_e}\right), \tag{6}$$

where Q_{raw} is the observed (uncorrected) cluster charge, E_e^{truth} is the true main electron energy of the interaction, F_0 is the charge-to-energy correction factor, t^{drift} is the true drift time of the

electron, and τ_e is the electron lifetime measured here independently and cross-checked against the simulation input.

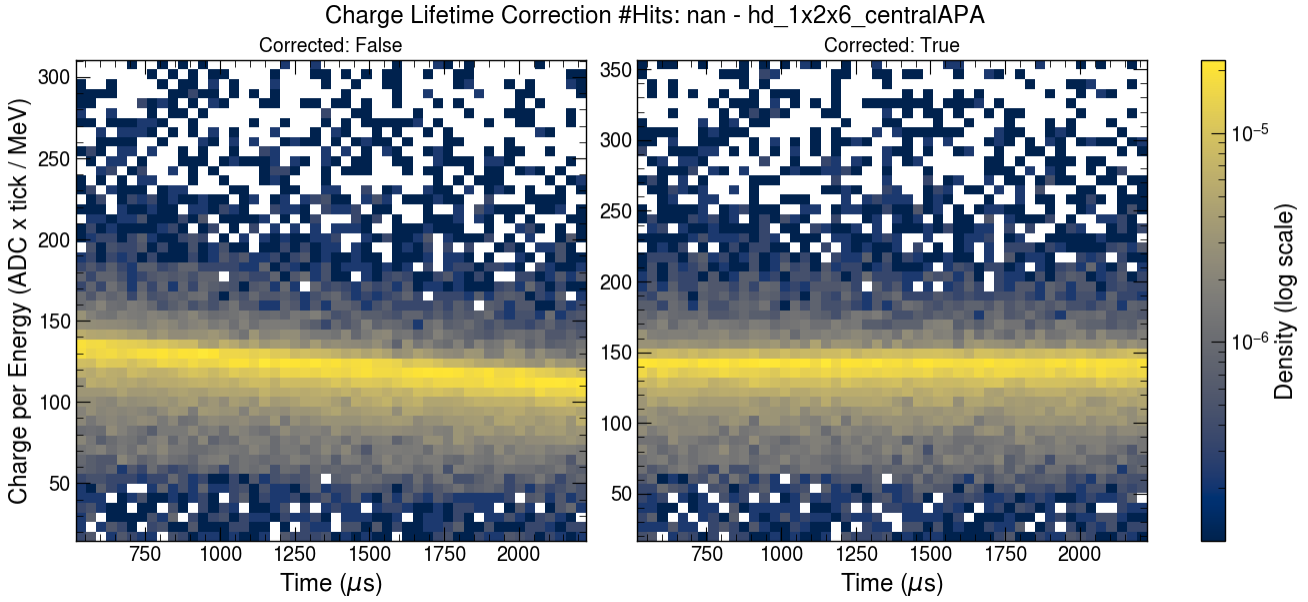


Figure 16: Drift attenuation correction.

Assuming the main cluster corresponds to a fully-contained electron, we take its known true energy (from simulation) as a reference and compute the charge-to-energy conversion factor for each (n_{hit}) threshold. Additionally, to assess the impact of this assumption, with the goal of characterizing signal clusters in the context of background which limits the clustering parameters, one can also investigate a "cheated charge". This is computed for all clusters, adding the surrounding charge that has been backtracked to the main electron of the interaction. A calibration based on this cheated clustering gives us an understanding of the losses that arise from a limited clustering ability in the context of background.

$$E_{\text{cluster}} = \alpha(n_{\text{hit}}, E_{\text{true}}) \cdot Q_{\text{corr}}. \tag{7}$$

After the truth drift correction has been applied to the data, the charge correction factor is immediately obtained.

Figure 17 shows the charge correction factor distributions for both the main cluster and the cheated electron cluster. The correction factor varies with increasing number of hits. Due to the nature of clustering, the first two points of the distribution are affected by threshold effects and present a high charge-to-energy ratio to compensate for the lack of completeness. Intermediate hit clusters have the lowest charge-to-energy ratio, as they are more likely to be fully contained within a few hits. As the number of hits increases, the clustering efficiency decreases, leading to a higher charge-to-energy ratio. This is because higher energy electrons produce more extensive ionization tracks, which are more challenging to fully capture within a single cluster.

This trend is well-described by an exponential plus logistic function. The functional form used to fit the charge correction factor as a function of the number of hits is given by:

$$\alpha(n_{\text{hit}}) = a \cdot \exp(-b \cdot n_{\text{hit}}) + c \cdot (1 + \exp(-d \cdot n_{\text{hit}}))^{-1}, \tag{8}$$

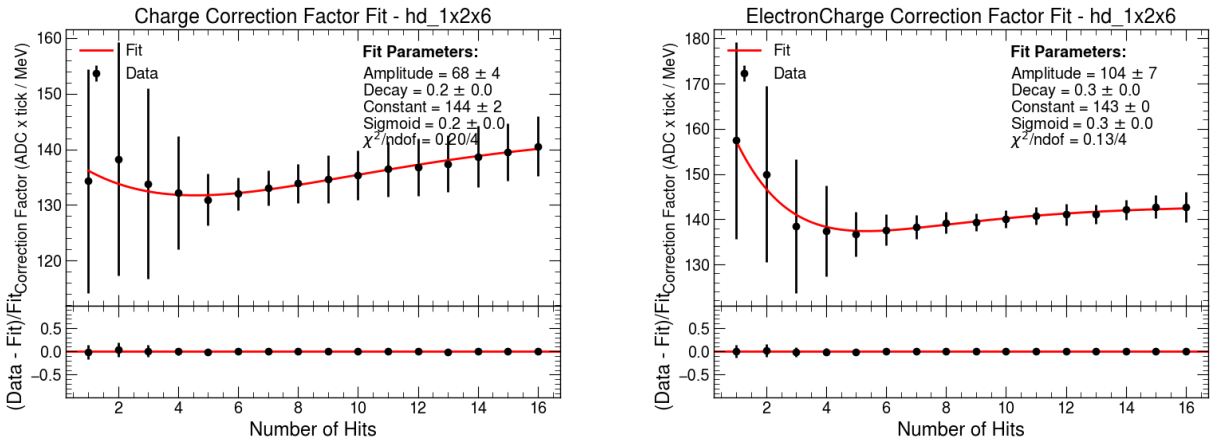


Figure 17: Per hit correction factor distributions. (left) Primary cluster charge. (right) Cheated electron cluster charge.

where a, b, c, d are fit's exponential amplitude, decay, limiting constant, and sigmoid growth parameters, respectively. This function captures the initial rapid decrease in the correction factor for low hit counts, followed by a more gradual increase at higher hit counts, reflecting the underlying physics of charge collection and clustering in the detector.

4.2 Electron Energy Calibration

Even after lifetime and charge calibration, small systematic biases may remain in the reconstructed electron energy due to recombination modelling, imperfect response, or clustering inefficiencies. We apply a linear correction to account for this residual offset:

$$E_e^{\text{final}} = \left(E_{\text{cluster}} - n\right)/m, \tag{9}$$

where m and n are derived from the linear fits to the simulated E_{true} versus E_{reco} distributions as shown in fig. 18. The calibration is performed separately for the main electron cluster and the cheated electron cluster. The calibration factor is computed as a function of the number of hits in the cluster (see **??**), which serves as a proxy for the electron energy.

As shown in fig. 19, the calibrated parameters approach unity for m and continuously rise for n as the number of hits increases, indicating worse electron clustering for higher–energy particles. There is no significant difference between the HD and VD geometries in terms of these calibration parameters.

4.3 Adjacent Cluster Calibration

Applying this calibration to the main electron cluster, we obtain the final calibrated energy of the main electron. For all the adjacent clusters, we simply apply the average charge correction factor obtained from the main electron calibration. There is no additional calibration applied to the adjacent clusters, as the charge calibration factor from the main electron cluster is used. However, we observe a small bias in the reconstructed gamma energy, which is due to the

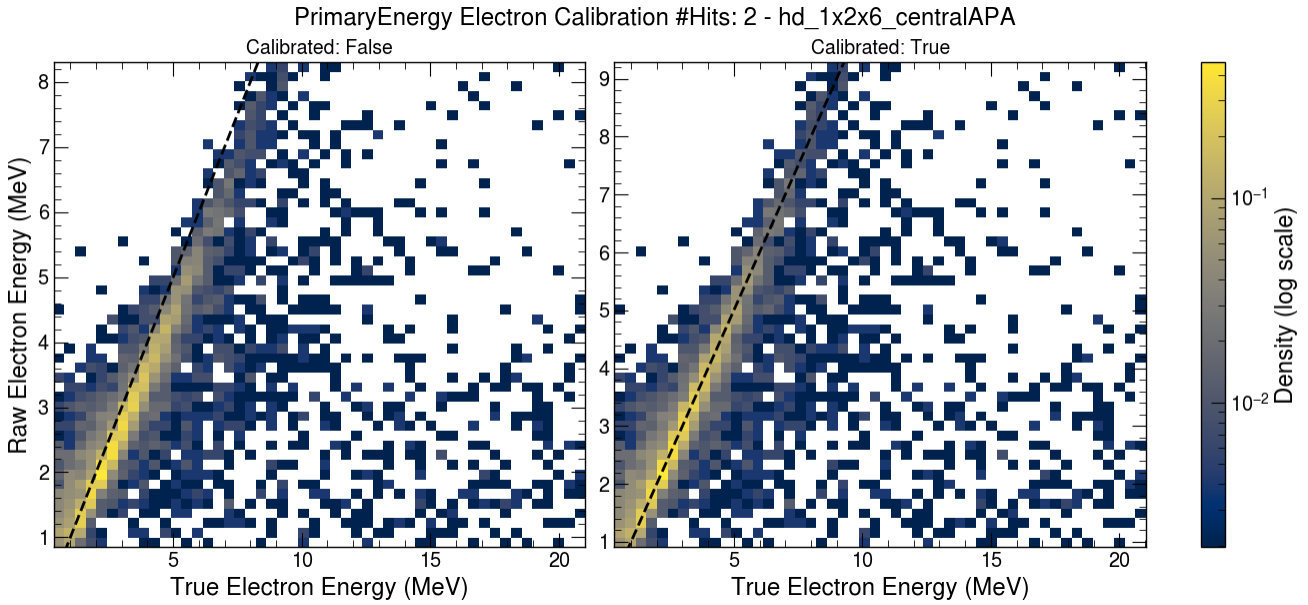


Figure 18: Charge calibration for the main electron clusters. (left) Comparison between true and reconstructed energy before linear fit calibration. (right) Comparison between true and reconstructed energy after linear fit calibration. Dashed line shows the values for which $E_{\text{reco}} = E_{\text{true}}$ as reference.

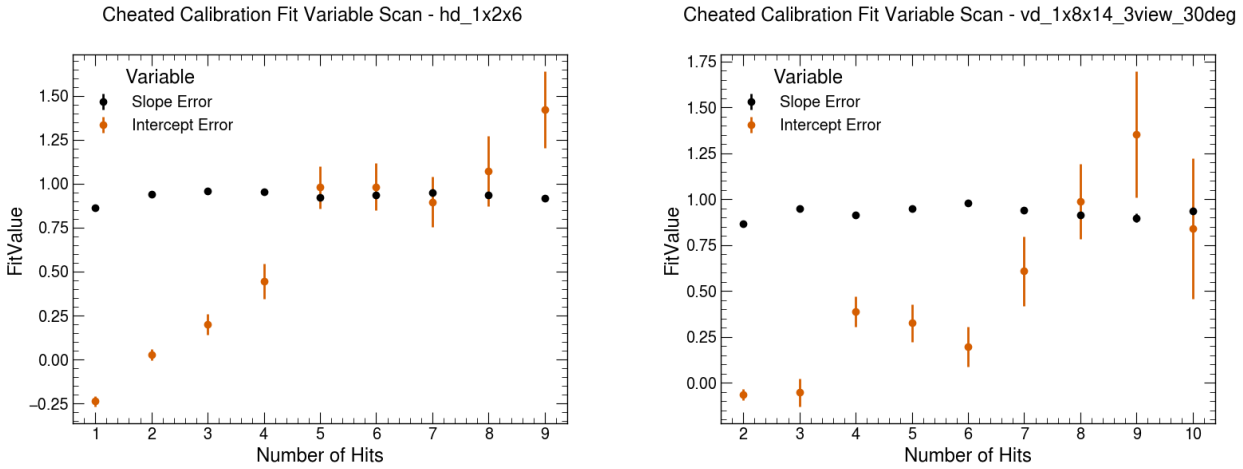


Figure 19: Calibration parameters as a function of number of hits in the cluster for (left) HD and (right) VD.

clustering inefficiency and the low energy of the gammas. This bias leads to a poor identification of the nuclear transitions in the neutrino energy spectrum. In fig. 20, we show the reconstructed gamma energy distributions for the different nuclear transition types: Gamow-Teller and Fermi. The Fermi transition has a single emission line at 4.38 MeV, while the Gamow-Teller has several.

The bias in the reconstructed gamma energy could be corrected by applying a linear calibration, similar to the one applied to the main electron cluster. However, due to the low energy of the gammas, the additional presence of background and the clustering inefficiency of the main electron, it has been considered unnecessary to apply any additional calibration to the adjacent clusters and instead focus on improving the channel selection and clustering efficiency without further bias. Additional work needs to be performed in order to estimate the reconstruction potential that can be obtained from correctly tagging these nuclear transitions.

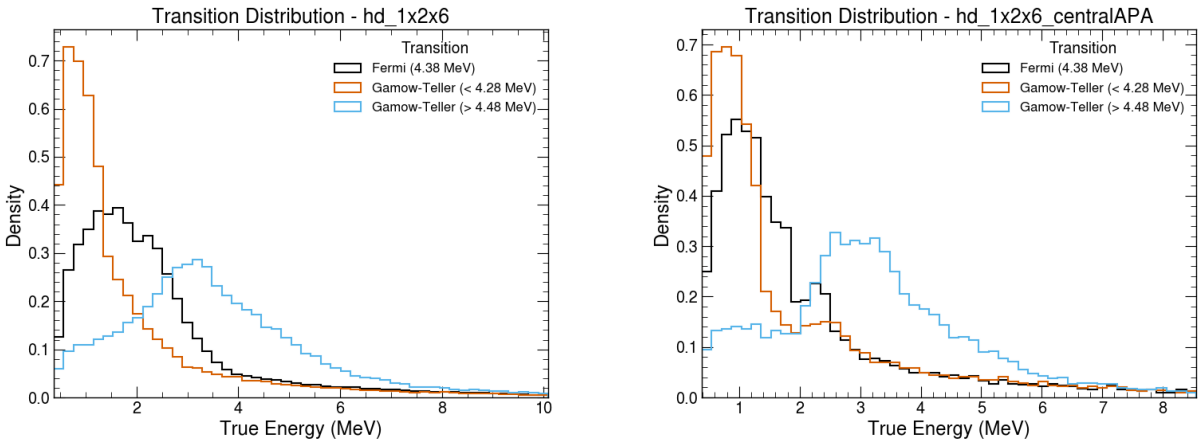


Figure 20: Adjacent cluster gamma signal (left) compared to the selected adjacent clusters (right) for the HD productions with and without background activity.

When considering the presence of background activity, as shown in fig. 20, the identification of the nuclear transitions becomes even more challenging. The background activity introduces additional low-energy deposits that can be misidentified as gammas from the neutrino interaction, further smearing the reconstructed gamma energy spectrum. This highlights the importance of effective background mitigation strategies and advanced reconstruction techniques to accurately identify and reconstruct the gamma energies associated with neutrino interactions in a realistic detector environment.

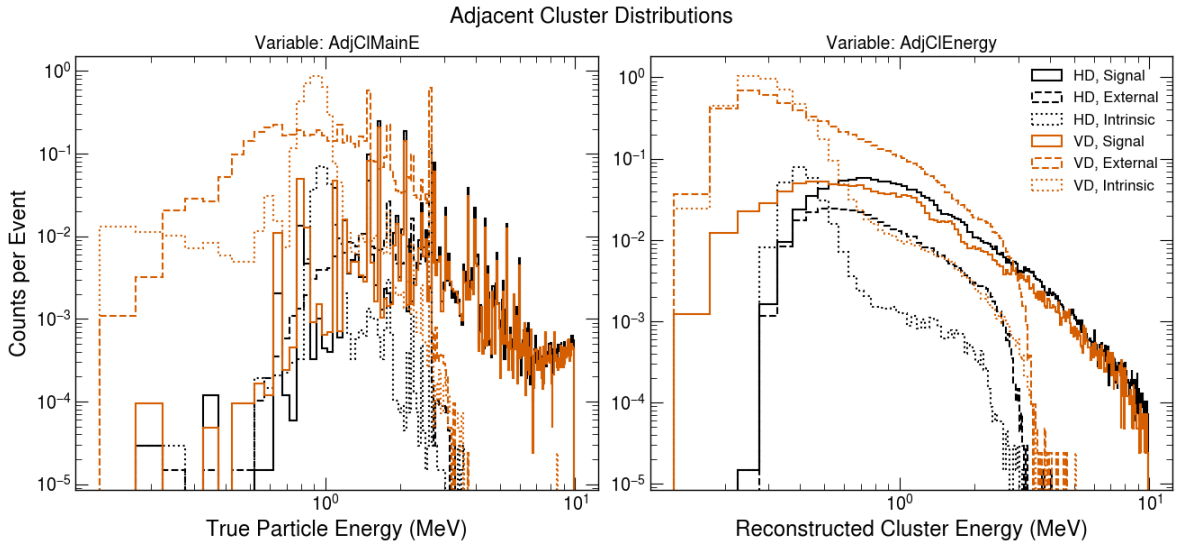


Figure 21: Adjacent cluster true (left) and reconstructed (right) energy distributions in the HD and VD geometries.

Another way to illustrate the challenge of reconstructing the gamma energies in the presence of background is to look at the true energy distributions of the adjacent clusters for different background scenarios. Figure 22 shows how the presence of background activity significantly increases the number of low–energy adjacent clusters (<3 MeV), which can be misidentified as gammas from the neutrino interaction. From this comparison it is clear that the shielded background scenario reduces considerably the amount of low–energy deposits, but this effect becomes weaker at higher energies.

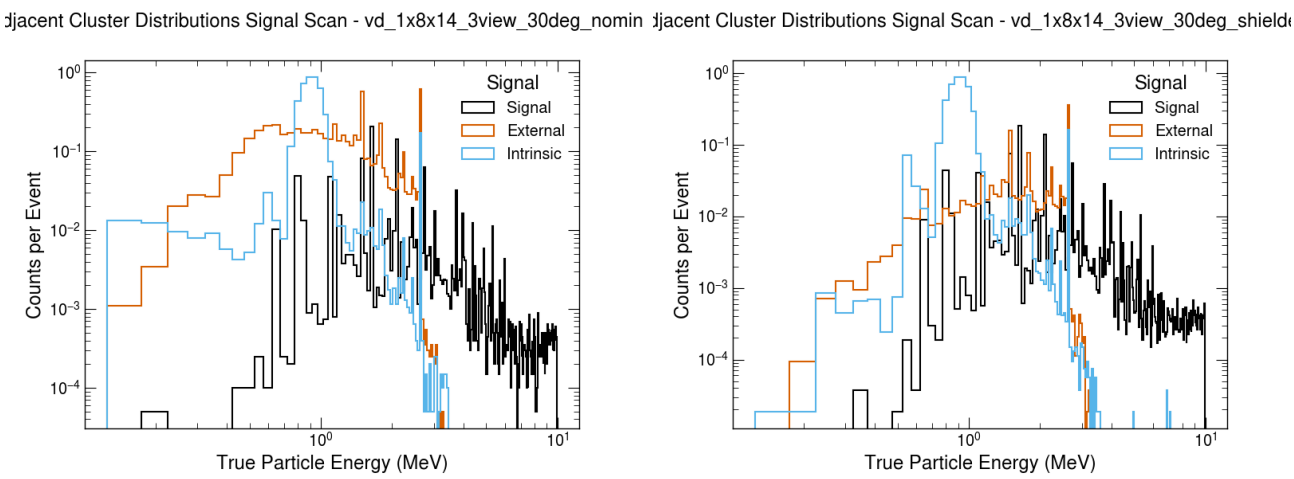


Figure 22: Adjacent cluster true energy distributions for the nominal (left) and shielded (right) background scenarios in the VD geometry. Notice the decrease in low–energy deposits (<1 MeV) when using the shielded background.

4.4 Neutrino Energy Calibration

The main electron energy is corrected for the drift time attenuation and the charge calibration factor, as described in section 4.2. The main electron energy is then used to compute the total neutrino energy, which is computed from the main electron energy and the energies of all adjacent clusters (up to 1 m). The adjacent clusters are assumed to be gamma-induced "blips" and are calibrated using the same charge calibration factor as the main electron. The total neutrino energy is computed as:

$$E_{\nu}^{\text{raw}} = E_{e}^{\text{final}} + \sum_{i} E_{\gamma,i}^{\text{final}}. \tag{10}$$

This strategy is the first and most obvious approach for neutrino energy reconstruction (**TotalEnergy**). However, it does not account for the energy smearing of the signal, and the presence of background can significantly affect the reconstructed neutrino energy. Therefore, we study alternative methods to improve the energy resolution and mitigate the effects of background contamination.

The next simple solution is to apply a selection on the adjacent clusters (**SelectedEnergy**), which are assumed to be gamma-induced "blips". The selection is based on the adjacent cluster charge and distance to the main electron cluster. The adjacent clusters are selected if they have a charge above a certain threshold and are within a certain distance to the main electron cluster. The thresholds are tuned to maximize the signal over background ratio. fig. 23 shows the distribution of the adjacent clusters charge and distance to the main electron cluster and selection thresholds.

Table 7 summarizes the different thresholds used for the adjacent cluster selection, based on the background scenario. The selected adjacent clusters are then summed to the primary cluster energy to obtain the reconstructed neutrino energy (**SelectedEnergy**).

The next step is to apply a machine learning algorithm to the adjacent clusters to improve the energy resolution. The algorithm is trained on the main cluster properties and the average adjacent cluster features. The algorithm is a random forest regressor, which is trained on the simulated data. The random forest regressor is used to predict the energy of the adjacent

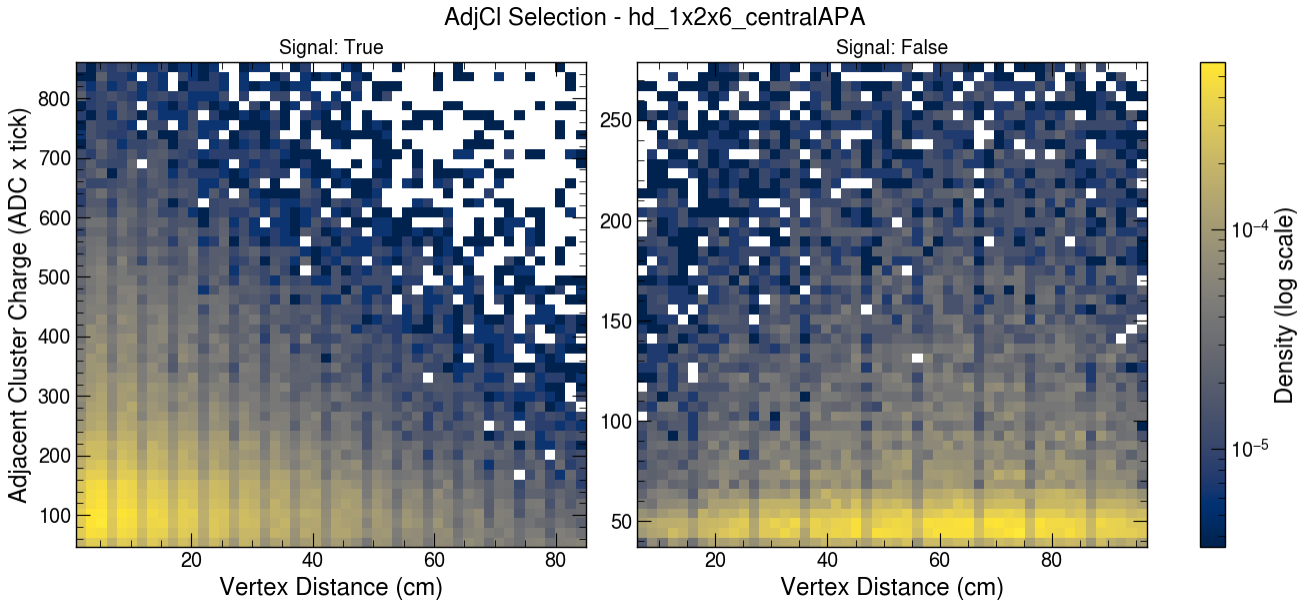


Figure 23: Distribution of the adjacent clusters charge and distance to the main electron cluster.

Configuration	Charge (ADC x tick)	Distance (cm)
hd_1x2x6	5	50
hd_1x2x6_centralAPA	10	30
hd_1x2x6_lateralAPA	8	40
vd_1x8x14_3view_30deg	12	25
vd_1x8x14_3view_30deg_nominal	15	20
vd_1x8x14_3view_30deg_shielded	10	30

Table 7: Adjacent cluster selection thresholds for different background scenarios.

clusters, which are then added to the main electron energy to obtain the total neutrino energy (**SolarEnergy**). To simply the problem and motivated by the two main calibration lines from the CC interaction displayed in fig. 5, the Machine Learning (ML) algorithm is trained two add one of two total adjacent cluster energy values: 4.47 MeV and 6.41 MeV.The score of the random forest regressor which provides an AUC of 0.77 for the test sample, which means that the algorithm is able to distinguish between signal and background with a good efficiency. The random forest regressor is trained on the following features:

- NHits: number of hits in the main electron cluster.
- AdjClNum: number of total adjacent clusters.
- SelectedAdjClNum: number of selected adjacent clusters.
- Energy: energy of the main clusters.
- **TotalAdjClEnergy**: energy of the total adjacent clusters.
- SelectedAdjClEnergy: energy of the selected adjacent clusters.
- SelectedAdjClEnergyRatio: energy ratio of the selected to the total adjacent clusters.
- **MaxAdjClEnergy**: maximum energy of the total adjacent clusters.

- SelectedMaxAdjClEnergy: maximum energy of the selected adjacent clusters.
- TotalEnergy: reconstructed **TotalEnergy**.
- SelectedEnergy: reconstructed **SelectedEnergy**.

From which the highlighted features are the most important for the random forest regressor, as shown in fig. 24. Finally, as a benchmark for comparison between the here presented reconstruction algorithms and with the aim of quantizing the impact of adjacent gammas on the total neutrino energy reconstruction, we introduce the **ClusterEnergy** algorithm as the reconstruction method based purely on the main electron cluster. The only correction is a fixed calibration offset applied to the full sample to align the center of the distribution with the true energy.

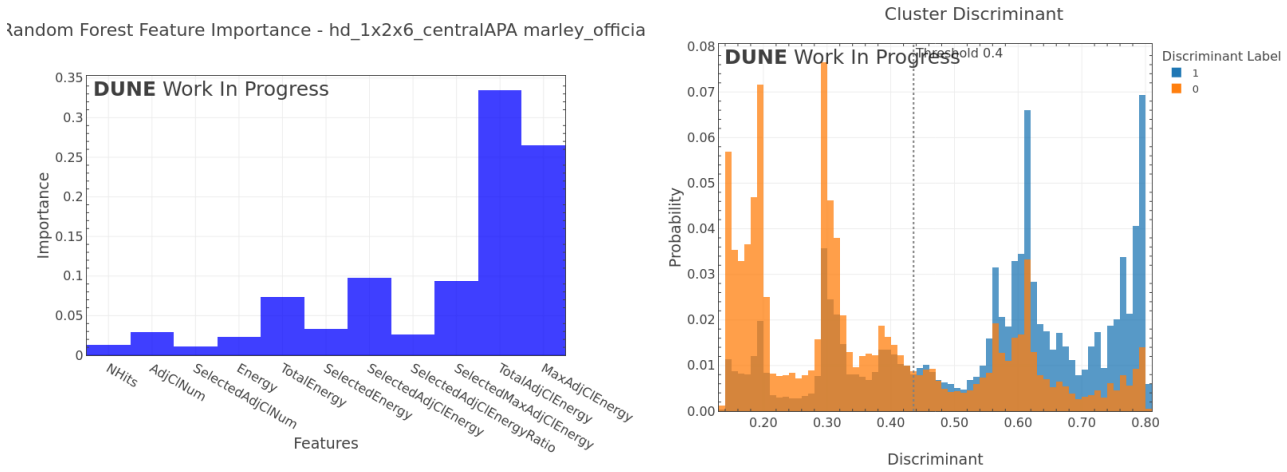


Figure 24: (left) Feature importance of the random forest regressor. (right) Discriminant calibration for the random forest regressor.

4.5 Neutrino Energy Smearing

Due to the missing nuclear binding energy and undetected soft particles, the reconstructed neutrino energy is expected to systematically underestimate the true value. As a final step to the reconstruction process, all energies (**TotalEnergy**, **SelectedEnergy**, **SolarEnergy** and **ClusterEnergy**) are calibrated with a linear fit to compensate for systematic losses in the previous clustering or calibration processes. This final global correction is applied to bring the distribution into agreement with the simulation truth:

$$E_\nu^{\text{cal}} = \left(E_\nu^{\text{raw}} - n\right)/m, \tag{11}$$

with m and n obtained from a linear fit to the true versus reconstructed neutrino energy spectrum of the MC sample as shown in fig. 25.

5 Results

From the TPC cluster vertex reconstruction and the flash–matching, we can extract the energy and vertex resolution. The calibrated neutrino energy is obtained from a standalone code

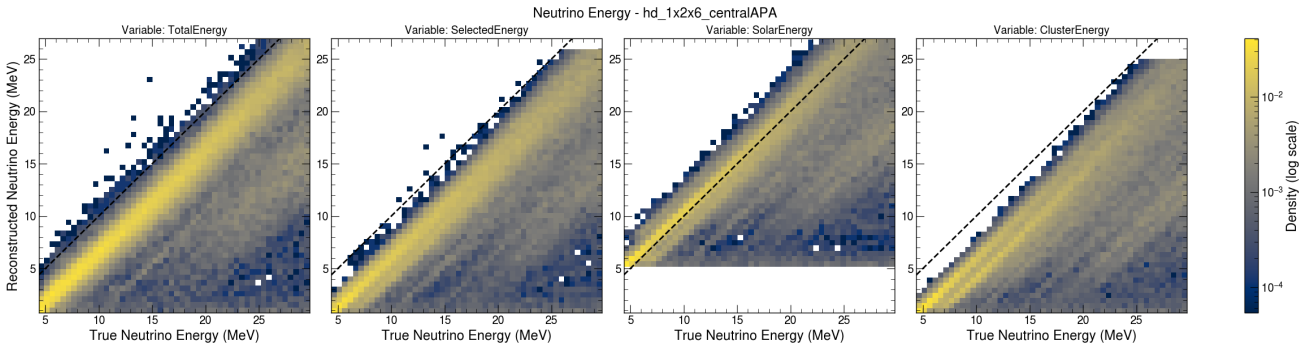


Figure 25: Neutrino energy smearing from the calibrated signal.

SOLAR package, which formats the simulation output into numpy arrays and performs the successive operations presented in section 3. Here we present the main figures of merit and compare the resolution performance between the two geometries, the *hd_1x2x6_centralAPA* and the *vd_1x8x15_3view_30deg* and additional background reduced scenarios.

5.1 Neutrino Vertex Smearing

As explained in section 3, the drift–coordinate of the reconstructed vertex is obtained from the flash–matching algorithm, which combines the information from the TPC cluster and the associated PDS flash. The results are shown in fig. 26 for the *vd_1x8x14_3view_30deg* geometry with nominal background production. These distributions are not Gaussian, as expected from the nature of the matching algorithms. The resulting matching can be correct, leading to a small error, or completely wrong, leading to a large error in the reconstructed coordinate.

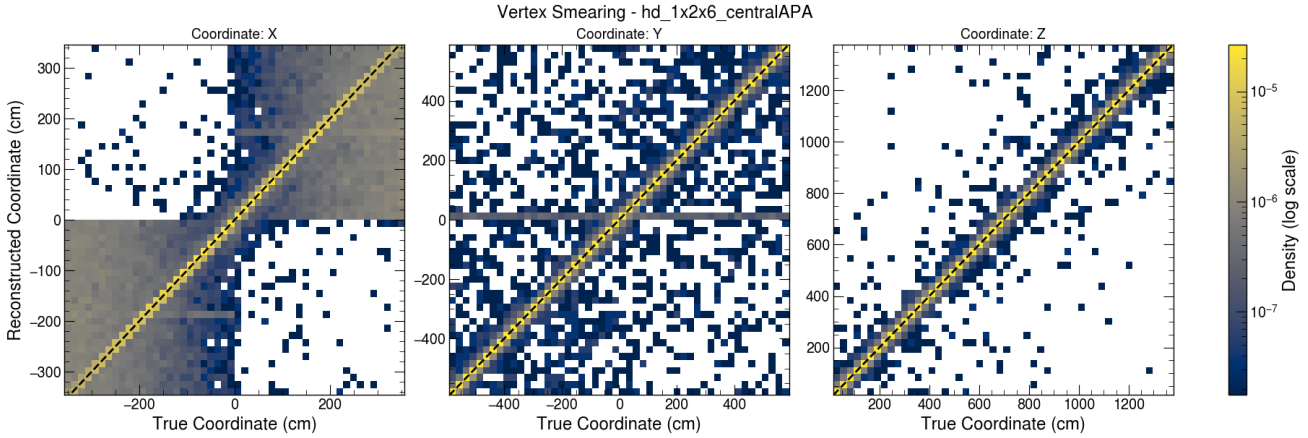


Figure 26: Vertex smearing from calibrated signal.

To have a better understanding of the reconstruction performance, we can compute the cumulative vertex error fig. 27. This plot shows the percentage of events that are reconstructed within a certain vertex error tolerance. For instance, we can see that approximately 90% of the events have a vertex error below 1 cm in the y– and z–coordinates, while only about 60% of the events achieve this level of accuracy in the x–coordinate.

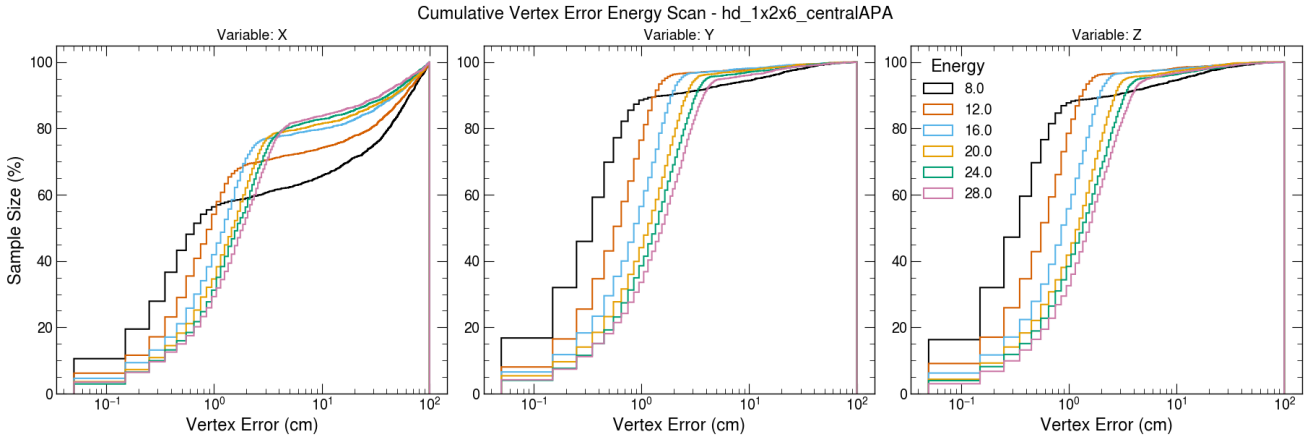


Figure 27: VD cumulative vertex error distribution.

5.2 Neutrino Vertex Resolution

The vertex resolution is defined from the distance between the reconstructed vertex and the true vertex position. Note that this analysis reconstructs particle interactions as point–like objects and therefore reconstructs a vertex that is shifted on average to the center of the true particle track. For the cases in which a coordinate cannot be determined (applicable to x– and y–coordinates), the reconstructed coordinate is set to the center of the active volume. The vertex error is then defined as the difference between the reconstructed and true coordinates.

Given that the distributions are not Gaussian, we can still attempt to fit a Gaussian function to the vertex error distributions by ignoring the tails to extract a vertex resolution from the FWHM of the fit as a figure of merit. This is done by fitting a Gaussian function to the core of the distribution. The results are shown in fig. 28 for the *vd_1x8x14_3view_30deg* geometry with and without nominal background production.

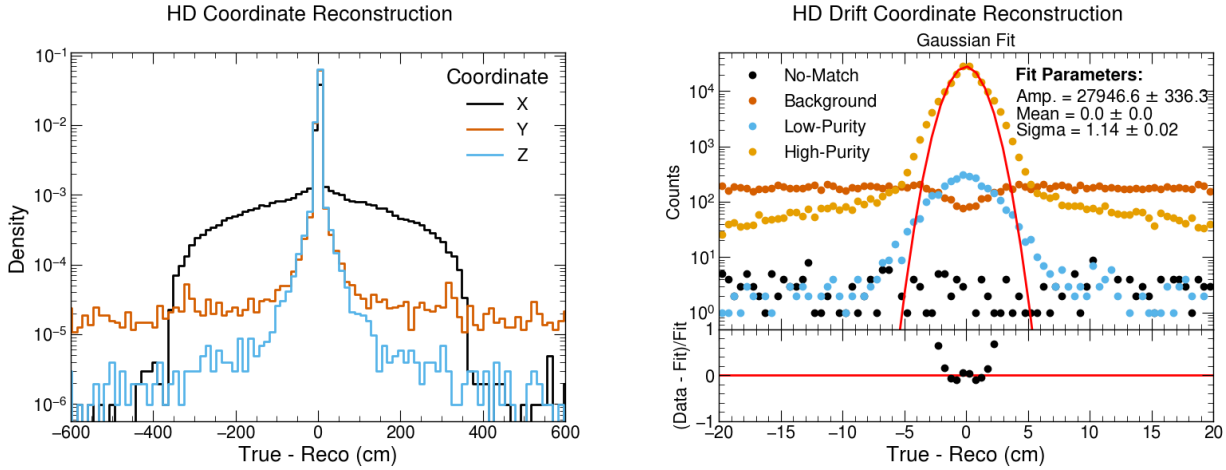


Figure 28: HD vertex error distributions. (left) All coordinate error distributions. (right) X coordinate error divided in high– and low–purity matched flash samples with core distribution fit.

The results of this fitting procedure as a function of true neutrino energy are presented in fig. 28 for both the neutrino-only and neutrino-plus-background samples. The vertex resolution is found to be between 0.5 cm and 2.5 cm except for the x–coordinate in the neutrino-plus-background sample, which is between 1.5 cm and 3.5 cm. This degradation in the x–coordinate resolution is

expected due to the pile–up of background events, which leads to a smearing of the time resolution of individual `ophits` in the correctly associated flashes. The degradation in vertex resolution with increased neutrino energy is a side effect of the point–like reconstruction approach, which places the reconstructed vertex at the center of the neutrino-induced particle track. As the neutrino energy increases, the average particle track size also increases, leading to a larger displacement between the true interaction vertex and the reconstructed vertex. Currently, there is no low–energy physics analysis in DUNE that would strongly benefit from a vertex resolution better than a few centimeters. For example, in the case of solar neutrinos, where external cavern–gammas are one of the major backgrounds, the Compton scattering length is on the order of tens of centimeters, which is much larger than the achieved vertex resolution. But, if needed, the vertex resolution could be improved by implementing a more sophisticated reconstruction algorithm that takes into account the topology of the event.

5.3 Fiducialization Efficiency

In order to reduce the impact of background events originating from the detector materials and the surrounding environment, a fiducial volume cut is typically applied to the reconstructed neutrino events in low energy–analyses. This cut is based on the reconstructed position of the primary TPC cluster associated with the neutrino interaction. The fiducial volume is defined as a region within the detector that is at least x cm away from any external detector boundary. This distance can be chosen in any analysis to optimize the signal–to–background ratio. The results for the HD geometry with central APA background scenario are shown in fig. 29 as a function of the distance to the detector boundary for different neutrino energies. The efficiency is calculated as the fraction of events with a reconstructed vertex position that lies within the fiducial volume, relative to the total number of simulated neutrino interactions occurring within the fiducial volume. As expected, the efficiency increases with increasing distance from the detector boundary, as events occurring closer to the boundary are more likely to be miss–reconstructed. Additionally, the efficiency improves with increasing neutrino energy, as higher energy events tend to produce more ionization charge and scintillation light, leading to better vertex reconstruction accuracy.

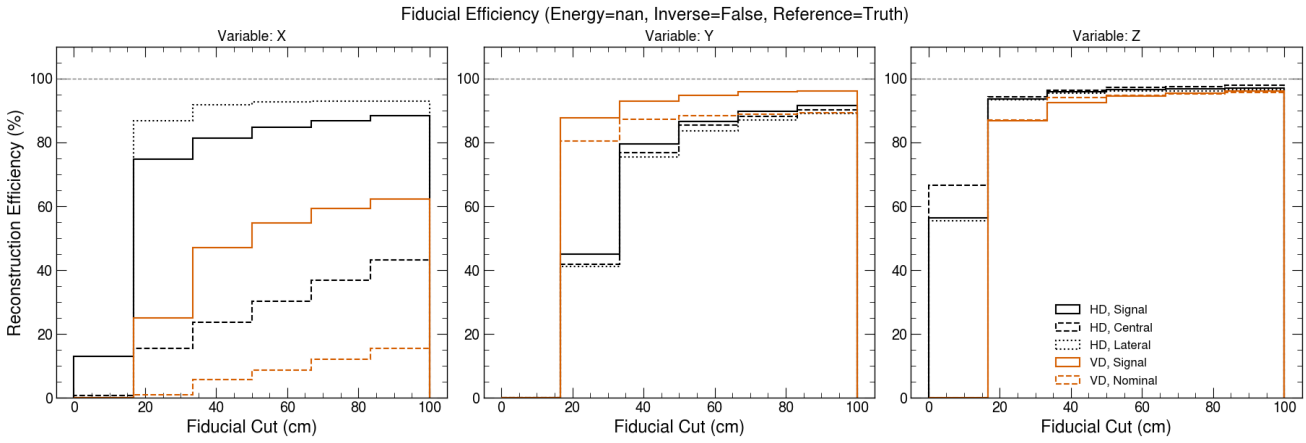


Figure 29: Fiducialisation efficiency as a function of cut–distance to the detector boundary for the average neutrino energy sample of 4 MeV to 30 MeV. From (left) to (right) X, Y and Z coordinates.

For a better comparison between the modules, we evaluate the average fiducialisation efficiency up to 100 cm for 10 MeV neutrino energies. The results for the HD and VD geometries are

shown in table 8, respectively. The efficiency values are generally high, especially in the y– and z–coordinates, where they exceed 65% in all cases. The x–coordinate shows a lower efficiency, particularly for the VD geometry, which can be attributed to the larger drift volume and lower TPC–PDS matching efficiency. The central HD volumes also have a lower fiducialisation capacity towards their edges, but thanks to the intrinsic fiducialisation provided by the lateral APAs,

Fiducialisation Efficiency (%) – 10 MeV			
Configuration	Coordinate		
	X	Y	Z
HD Signal	71.6 ± 0.7	65.7 ± 0.8	89 ± 3
HD Lateral	78 ± 1	65 ± 2	80 ± 2
HD Central	18.6 ± 0.6	65 ± 2	98 ± 9
VD Signal	36.1 ± 0.6	78.9 ± 1.0	76 ± 1
VD Nominal	4.5 ± 0.8	73 ± 4	78 ± 6

Table 8: Average vertex fiducialisation efficiency at 100 cm from the detector boundary.

5.4 Electron Energy Resolution

As a figure of merit for the process, we can compute the energy resolution from the calibrated energy of the main electron. To remove any reconstruction artifacts, this first result uses:

- **Ideal Clustering**. This means that all the hits associated with the electron are perfectly clustered together, and no hits from other particles are included. This is done using truth information to select the hits.
- **High Hit Threshold**. A minimum of 3 hits is required to form a cluster. This helps to reduce noise and ensures that only significant energy depositions are considered.
- **True Drift Correction**. The drift correction is applied using the true position of the interaction, rather than relying on reconstructed positions. This eliminates any errors that might arise from inaccuracies in the reconstruction process.- Three scenarios are considered: **True** drift correction, **Reco** drift correction (using flash-matching), and **Mean** drift correction (using the average drift time for the detector).

This is typically the value that gets compute when estimating the energy resolution of a TPC. It is obtained for an idealized clustering with ad hoc calorimetry and a high hit threshold (in this case we use 3 for a typical track). The energy resolution is computed as the Root Mean Square (RMS) of the relative error of the reconstructed variable versus the truth. The results are shown in **??**. The formula for the energy resolution fit is:

$$\Omega_{(E_{\text{true}})} = \sqrt{\left(\frac{p_2}{\sqrt{E_{\text{true}} - b}}\right)^2 + \left(\frac{p_1}{E_{\text{true}} - b}\right)^2 + p_0^2}, \tag{12}$$

where p_0, p_1, and p_2 are the parameters of the fit, and b is a constant offset and corresponds to the detection threshold. These parameters can be interpreted as the intrinsic resolution evaluated to 11(1)%, the stochastic term, and the noise term, respectively. This formula con be derived

from the standard parameterization of energy resolution in calorimetry, which typically includes contributions from stochastic processes, electronic noise, and constant terms.

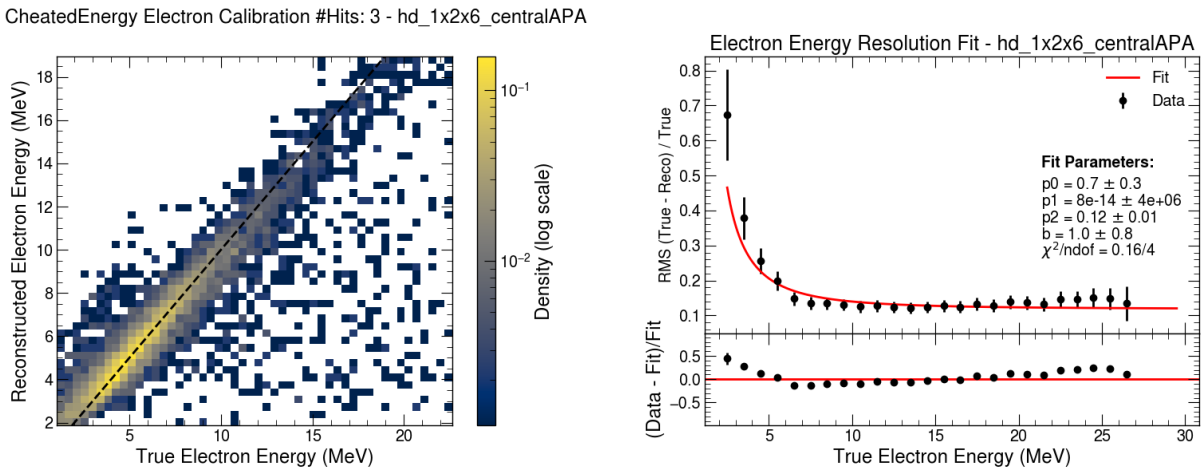


Figure 30: Energy resolution from calibrated charge with ideal clustering, true drift correction and hit threshold of 3. (Left) Heatmap of the reconstructed energies. (Right) Per bin resolution of the data.

To assess the impact of the different reconstruction stages on the energy resolution, we can vary one parameter at a time while keeping the others fixed. The results are shown in fig. 31.

As can be seen, the clustering has a significant impact on the energy resolution, especially with increasing energies. This is expected, as the clustering algorithm may not be able to perfectly reconstruct the electron, leading to a loss of energy.

The hit threshold also has an impact on the energy resolution, but it is less significant than the clustering. A higher hit threshold leads to a better energy resolution, as it selects electrons with higher completeness.

Finally, the drift correction has a small but noticeable impact on the energy resolution. Using the true drift correction leads to a better energy resolution although the reconstructed drift correction (using flash-matching) performs almost equivalently. The mean drift correction, on the other hand, leads to a degradation that is especially significant for the VD geometry. This is expected, as the mean drift error will be larger for larger detectors, leading to a larger uncertainty in the energy reconstruction.

5.5 Neutrino Energy Resolution

The energy resolution is defined as the ratio of the standard deviation of the reconstructed neutrino energy to the mean value of the neutrino energy. The energy resolution is calculated from the calibrated neutrino energies. The results are shown in fig. 32 for the 4 different reconstruction algorithms.

As expected, the **ClusterEnergy** algorithm has the worst energy resolution for neutrino energies below 10 MeV, as it does not take into account the energy deposited by adjacent gamma blips which can amount to more than 50% of the initial neutrino energy. The **SelectedEnergy** and **SolarEnergy** algorithms have the best resolution in this range. As the energy of the neutrino goes higher, the relative weight of the adjacent background clusters starts becoming less

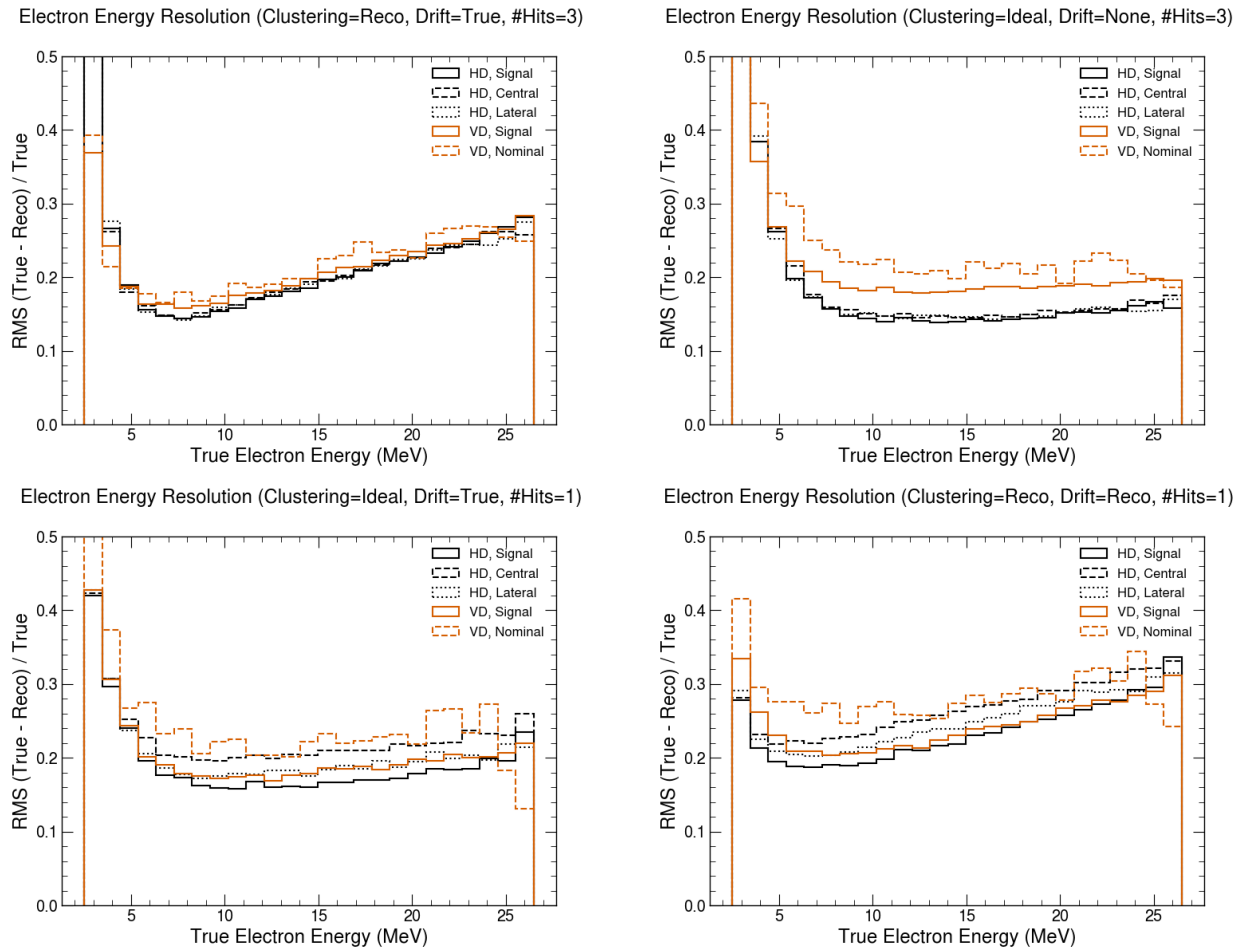


Figure 31: Effect of clustering (ideal vs reconstructed), drift correction (ideal vs reconstructed vs none), and hit threshold on the electron energy resolution. (Top Left) Shows the effect of non-ideal clustering. (Top Right) Shows the effect of not applying a drift correction. (Bottom Left) Shows the effect of lower hit threshold. (Bottom Right) Shows the final and complete electron energy resolution with reconstructed clustering and drift correction, as well as the lowest possible hit threshold.

important. Over the range of 10 MeV and 20 MeV all algorithms perform almost identical and for the 20 MeV and 30 MeV neutrino energies, the **TotalEnergy** algorithm starts to outperform the others. Importantly, this trend is maintained for any hit threshold selected, ensuring that the energy resolution is robust against the clustering criteria.

Overall, the **SolarEnergy** algorithm is the best choice for neutrino energies below 20 MeV, as it has better (or compatible) energy resolution compared to the other algorithms. The benefit of this method that is not shown in this comparison is that it sets an upper limit to the reconstructed energies of background particles. If not constrained, cavern backgrounds such as neutrons or gammas can contribute significantly with fake events to physics analyses such as the solar neutrino oscillation studies. By only adding a maximum fixed amount of energy to the main clusters produced by these particles, one ensures that their contribution is limited to lower energies compared to the full scope of the solar neutrino spectrum.

As already explained, the flash–matching step in the reconstruction process allows to associate a TPC clusters with the corresponding PDS flash and hence obtain the correct drift coordinate of the interaction. In the context of energy reconstruction, this has an impact on the drift attenuation correction on a cluster-by-cluster basis.

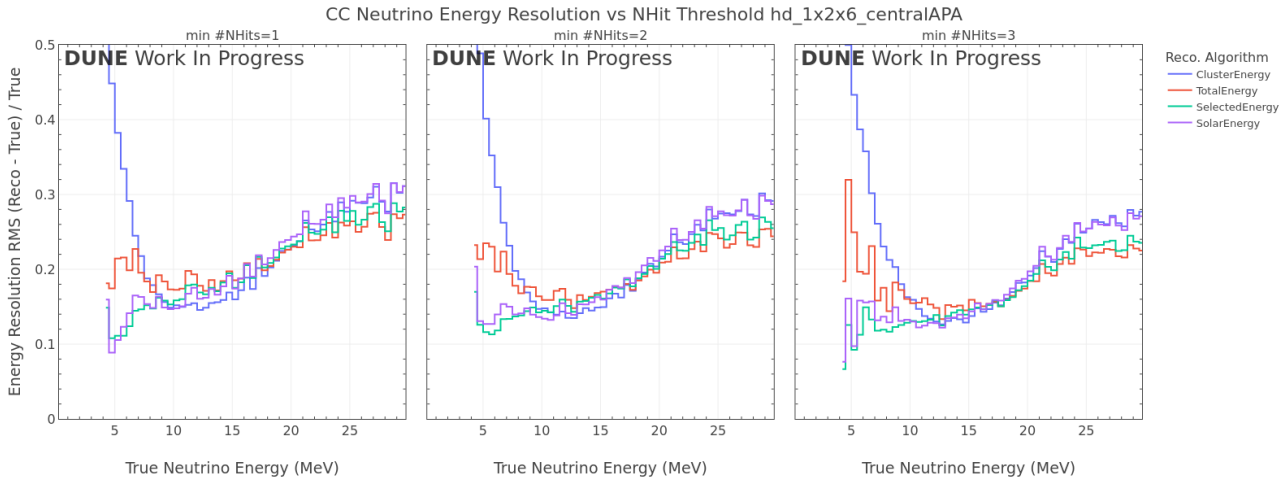


Figure 32: Energy resolution from calibrated neutrino energies with different reconstruction algorithms.

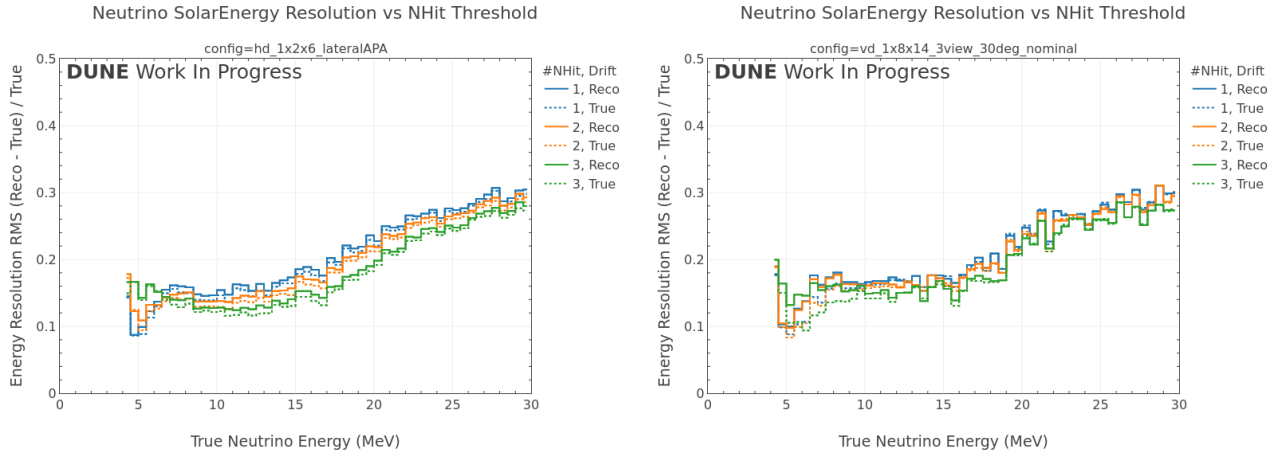


Figure 33: (left) HD (right) VD. Neutrino energy resolution with and without flash–matching. The dashed lines represent the ideal case where the drift coordinate is taken from truth information.

As can be seen, in fig. 33, the flash–matching step has a residual effect on the energy resolution. The difference between the flash–matched and non-flash–matched cases are small, indicating that the inefficiencies in the energy reconstruction are dominated by other factors, such as clustering inefficiencies and intrinsic energy losses due to nuclear effects and undetected particles. The dashed lines represent the ideal case where the drift coordinate is taken from truth information, showing that even in this idealized scenario, the resolution does not significantly improve.

6 Conclusion

In this technical note, we have presented the reconstruction algorithms for the DUNE FD, focusing on the neutrino energy and vertex resolution. The reconstruction is based on the TPC cluster vertex reconstruction and flash–matching, which allows us to extract the calibrated neutrino energy and vertex position.

6.1 Far Detector Comparison

So far we have used as reference the *hd_1x2x6_centralAPA* geometry, or compared between the *hd_1x2x6_lateralAPA* and the *vd_1x8x14_3view_30deg_nominal*. In this section, we compare all geometry and background configurations which are described in section 2.1.

In the case of the vertex resolution, we can use the previously defined figure of merit and quote the percent of events with a vertex error below a certain threshold. The results are shown in table 4.

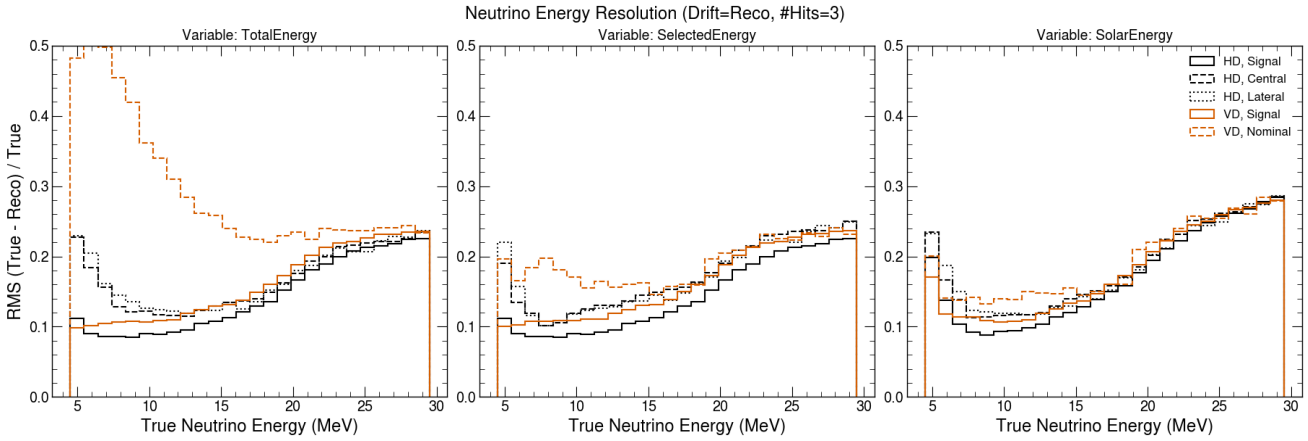


Figure 34: (left) **TotalEnergy** resolution comparison. (center) **SelectedEnergy** resolution comparison. (right) **SolarEnergy** resolution comparison between geometries.

Applying the **TotalEnergy** reconstruction algorithm, we can see that the energy resolution is better in the hd_1x2x6_centralAPA geometry, which is expected since it has a lower background rate and smaller drift volume causing therefore less diffusion. The energy resolution is shown in fig. 34 (left).

Applying the **SelectedEnergy** reconstruction algorithm, we can see that the energy resolution is again better in the HD geometry for the same reasons as before. But this time, the difference is less pronounced due to the fact that the **SelectedEnergy** algorithm maximizes the adjacent cluster significance.

Finally, applying the **SolarEnergy** reconstruction algorithm and the appropriate drift correction, we obtain the most similar results. This comes at the expense of a slightly worse energy resolution above 20 MeV.

We can see that the energy resolution is similar between the *hd_1x2x6_centralAPA* and *vd_1x8x14_3view_30de* geometries, with a slight advantage for the former. The *vd_1x8x14_3view_30deg_nominal* geometry has a worse energy resolution due to the higher background rate, but it is still acceptable for low-energy neutrino physics analyses.

6.2 Summary

The vertex and energy reconstruction of low–energy neutrino interactions in DUNE has been studied. The vertex resolution was evaluated using a point–like reconstruction approach that combines the TPC cluster vertex with the associated PDS flash. The achieved vertex resolution is between 0.5 cm and 3.5 cm, depending on the coordinate and the presence of background

events. There are significant differences in fiducialisation performance between the HD and VD geometries, with the HD geometry achieving higher fiducialisation efficiency due to its lower background rates. The energy reconstruction was performed using four different algorithms to sum the energy of reconstructed TPC clusters. The performance of these algorithms was evaluated in terms of energy resolution, fiducialisation efficiency, and the impact of flash–matching on the energy reconstruction. Contrary to the vertex resolution, the energy resolution does not show significant differences between the two geometries provided that one chooses the appropriate reconstruction algorithm. The best energy resolution is achieved using the *SolarEnergy* algorithm, which yields a resolution of approximately 20% at 15 MeV and improves to about 10% at 5 MeV. The flash–matching step has a minor impact on the energy resolution, indicating that other factors, such as clustering inefficiencies and intrinsic energy losses, dominate the resolution. Overall, the reconstruction algorithms demonstrate good performance for low–energy neutrino interactions in DUNE, with potential for further optimization and improvement. Based on the amount of background activity in the VD module, other reconstruction objectives such as channel tagging (out of the scope of this work) and particle identification are likely to be more challenging than in the HD module. The case of a shielded VD module, which has background rates comparable to the HD module, is therefore highly encouraged and expected to yield similar low–energy reconstruction performance.

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Acronyms

APA Anode Plane Assembly. 4, 5, 6, 7, 16, 26, 37, 38

CC Charged Current. 3, 5, 8, 10, 16, 18, 24, 33

CPA Cathode Plane Assembly. 4, 5, 6, 7, 26

CRP Charge Readout Plane. 5, 7

DUNE Deep Underground Neutrino Experiment. 1, 2, 3, 4, 5, 7, 8, 9, 10, 11, 12, 13, 14, 24, 41

ES Elastic Scattering. 8, 10, 24

FD Far Detector. 2, 3, 4, 5, 6, 24, 41, 42

Fermilab Fermi National Accelerator Laboratory. 4

Geant4 Geometry and Tracking v.4. 5, 8, 11

HD Horizontal-Drift. 4, 5, 6, 7, 13, 14, 17, 18, 20, 21, 23, 25, 26, 29, 30, 31, 36, 37, 38, 41, 42, 43

LAr Liquid Argon. 4, 5, 8, 9, 10, 11, 12, 13, 18, 21, 22, 25, 27

LArSoft Liquid Argon Software Framework. 5, 8, 11, 12, 14

LArTPC Liquid Argon Time Projection Chamber. 2, 3, 5, 12

MC Monte Carlo. 5, 8, 9, 14, 34

MIP Minimum Ionizing Particle. 13

ML Machine Learning. 33

NC Neutral Current. 8

PCB Printed Circuit Board. 5

PD Photon Detector. 13

PDE Photon Detection Efficiency. 13

PDS Photon Detection System. 2, 3, 4, 5, 12, 13, 14, 15, 16, 18, 19, 20, 21, 22, 27, 35, 40

PE Photoelectron. 13, 15, 18, 19, 21, 22, 23

RMS Root Mean Square. 38

ROI Region Of Interest. 12, 13, 14

SiPM Silicon Photomultiplier. 13

SURF Sanford Underground Research Facility. 4, 10

TPC Time Projection Chamber. 2, 4, 5, 6, 10, 12, 13, 14, 15, 16, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 35, 37, 38, 40, 43

VD Vertical-Drift. 4, 5, 6, 7, 13, 18, 19, 20, 21, 22, 23, 25, 26, 29, 30, 31, 32, 36, 37, 38, 39, 41, 43

XA X-ARAPUCA. 5, 12, 13, 19

Glossary

Decay0 is a program written in the early 90's by Volodymyr Tretyak and collaborators, using the Fortran 77 programming language and the CERNLIB library. It has been maintained and improved up to 2020. Later, a pure C++ version was released without any dependency on Fortran and CERNLIB. This code was integrated in 2011 as the genbb_help module of the Bayeux C++ library (version ¡ 4). BxDecay0 is a standalone library with very few dependencies (mostly the GSL library for numerical integration and a few special math functions). External random engines can be used through a simple wrapping functor interface, particularly the ones provided by the C++ standard random library can be used by default. However the user is free to provide its own uniform deviates random generator (based on GSL, ROOT or whatever).. 5, 8

MARLEY (Model of Argon Reaction Low Energy Yields) is a neutrino event generator designed to simulate low-energy neutrino interactions with argon nuclei. It is developed by the Low-Energy Neutrino group of the DUNE collaboration. MARLEY focuses on simulating interactions relevant to solar and supernova neutrinos in liquid argon detectors, providing detailed modeling of the nuclear physics involved in these interactions. It incorporates a gamma-decay scheme to describe the de-excitation of the residual nucleus after a neutrino interaction, which is crucial for accurately simulating the resulting signals in the detector. MARLEY is implemented in C++ and is designed to be integrated with the LArSoft framework used by the DUNE collaboration for detector simulation and event reconstruction.. 5, 8, 11, 24

7 Appendix

Introduction

Due to its nature as a noble gas, ^{222}Rn is difficult to remove from the detector environment, making it a persistent background source also in detector components. Among its progeny, several isotopes are particularly important:

- **Bismuth (^{214}Bi)**: A short–lived beta emitter with a high Q–value (up to 3.27 MeV) and multiple gamma lines. It is the most significant source of high–energy background from ^{222}Rn [13].
- **Bismuth (^{210}Bi)**: A beta emitter that produces a continuum low–energy background up to 1.16 MeV.
- **Lead (^{210}Pb)**: A long–lived isotope ($t_{1/2}$ = 22.3 years) that accumulates on detector surfaces [14].
- **Polonium (^{210}Po)**: An alpha emitter with a half–life of 138 days. It can be used as a surface contamination proxy due to its distinct alpha signal [15].